TABLE CONTENT

ACKNOWLEDGEMENT ii

TABLE OF CONTENT iii

LIST OF FIGURE v

LIST OF TABLES V

CHAPTER ONE 1

1. INTRODUCTION 1

1.1 Background of the Study 1

1.2 Objectives of the project 2

1.3 Specific objectives 2

1.4 Significance of the Project 2

1.5 Research Framework 3

1.6 Definition of Terms 4

CHAPTER TWO 6

2. Slab Design 6

2.1 Classification of slab 7

2.2 Slab analysis and design 7

2.3 Design Load 14

2.4 Moment Adjustment 21

CHAPTER THREE 47

3.1 Design of Stair 47

CHAPTER FOUR 53

Beam and Column Design 53

4.1 Design Constant 53

4.1.1 Depth Determination 53

4.2 Design for Support Beam 55

4.3 Design on Span Beam 58

CHAPTER FIVE 62

5. COLUMN DESIG.....62
5.1 classification of column …….……..…… 62
5.1.1 analysis of design procedure ……..…… 62
5.2 material property…….…….. 64
5.3 Effective length …….……. 65
CHAPTER 6 …….……. 73
6. Foundation Design …… 73
6.1 Design of Isolated Footing ……..…… 73
6.1.1 Depth of Footing …… 73

CHAPTER ONE

INTRODUCTION

Background of the Study

Building structures are composed of architectural and structural parts. The structural part of the building supports the body of the building preventing it from any collapse or failure. Therefore, structural design involves the determination of the different sections of the skeletal part of the building to make it stable and sustainable throughout its design life.

The researchers wanted to design a Hotel building in KOMBOLCHA at WOLLO University. A structural design is executed in such a way that the building will remain fit with appropriate Gate degrees of reliability and in an economic way. It should sustain all the actions and influences during execution and use. Therefore, structural design focuses on structural safety and serviceability with due durability. The structural design is calculated based on the Ethiopian Building Code of Standard (EBCS) prepared in 2015 E.C. This code follows the Limit State design approach. Limit state is a state beyond which the structure is no longer satisfies the design performance requirements. It consists of two states namely Ultimate Limit and serviceability Limit states. Ultimate Limit states are conditions related with collapse or states prior to structural failure. Its main concern is the safety of structure and people. Serviceability Limit states are those associated to conditions beyond which a structure does not accomplish specified service requirements. It is mainly concerned about the function of construction works, comfort of people, and appearance. The prime objective of the design is structural safety and serviceability. In case if the structure fails, it must be in such a way it will minimize risks and casualty. It must extend the time for evacuation of people inside a building. This requirement of structural design is accomplished by the principle called ductility. Ductility allows yielding of steel reinforcement prior to the collapse of the building. Yielding of steel bars warns the start of failure of a structure or its part. Therefore, structures are designed to be under reinforced by certain percent to assure ductility mode of failure if it happens.

Reinforced concrete, as a composite material, has occupied a special place in the modern construction of different types of structures due to its several advantages. Italian architect Point once remarked that concrete liberated us from the rectangle. Due to its flexibility in form and superiority in performance, it has replaced, to a large extent, the earlier materials like stone, timber and steel. Further, architect's scope and imaginations have widened to a great extent due to its moulid ability and monolith city. Thus, it has helped the architects and

engineers to build several attractive shell forms and other curved structures. However, its role in several straightline structural forms like multistoried frames, bridges, foundations etc. is enormous.

The design of these modern reinforced concrete structures may appear to be highly complex. However, most of these structures are the assembly of several basic structural elements such as beams, columns, slabs, walls and foundations. Accordingly, the designer has to learn the design of these basic reinforced concrete elements. The joints and connections are then carefully developed.

Objectives of the project

The foremost objective of the project is to design a G+4 and terrace hotel building in KOMBOLCHA at WOLLO University

Specific objectives were:

1. To calculate the loads that are acting on the building such as dead loads, live loads, wind loads and earthquake forces based on the Ethiopian Building Code and Standards (EBCS).
2. To design the structural stability of the building structure members such as slab, beam, column, stair and footing foundation.
3. To design the building by ultimate state using manual calculation and computer software's.

Significance of the Project

After completing the project study, the results and findings will be helpful to the students taking up civil engineering course, that they could grasp the ability and knowledge of analyzing and designing a given structural building of any type using the methods and procedures stated therein.

Civil engineering students could have a proper knowledge of using different computer software's in engineering design. Civil engineering students could get an idea of the application of design theories in structural engineering to the actual design of buildings.

Research Framework

DESIGN OF G+4 HOTEL BUILDING

↓

PRELIMINARY PREPARATION OF ARCTICTURAL DRAWING

↓

DESIGN LOAD CALCULATION (DEAD

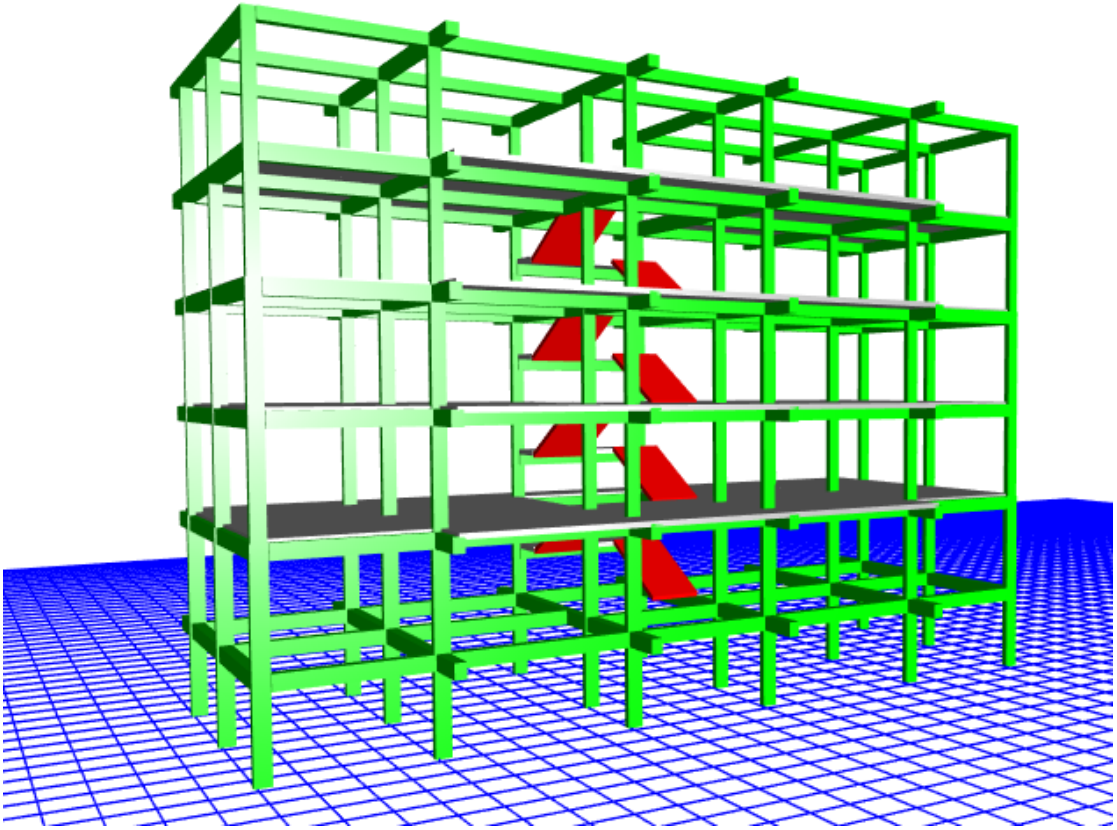
↓

STRUCTURAL DESIGN ANALYSIS (Slab, Beam, Column, Stair, Footing)

↓

PREPARATION OF STRUCTURAL DRAWING

Definition of Terms



Beams: are structural elements carrying external loads that causes bending moments, shear forces and torsion moments along their length. The beams can be; singly or doubly reinforced, and simply supported, fixed or continuous. The structural detail of such beams must resist bending, diagonal tension, shear and torsion, and must be such as to transmit forces through a bond without causing internal cracks.

Columns: are vertical members who transfer the load from beam to foundation, usually under compression and they are classified as short, long columns and slender columns. Long columns are liable to buckle under axial loads.

Foundation: is the part of the structure that is usually placed below the surface of the ground and that transmits the load to the underlying soil or rock. All soils compress noticeably when loaded and causes the supported structure to settle. The two essential requirements in the design of foundations are that the total settlement of the structure be limited to a tolerably small amount and the differential settlement of the various parts of the structure be eliminated as nearly as possible.

Joints- the connection of every building component could be in between beam & column, column & ground or beam with roof etc…

Slab - is a two dimensional structural member, it also called a surface element which transfers the loads to the beams or column in case of flat slab. The type of slab is depends on support condition, arrangement and load transfer mechanism etc. Slabs may be simply supported or continuous over one or more supports and are classified according to the method of supports as follows; spanning one-way between beams or walls, spanning two way b/n the support beams or walls and flat slabs carried on columns and edge beams or walls or with no interior beams.

Stair - is one-step in a flight of stairs. Also called a stair step. In buildings, stairs is a term applied to a complete flight of steps between two floors. A stair flight is a run of stairs or steps between floors. A staircase or stairway is one or more flights of stairs leading from one floor to another, and includes landings, newel posts, handrails, balustrades and additional parts. A stairwell is a compartment extending vertically through a building in which stairs are placed.

CHAPTER TWO

Slab Design

Slabs are horizontal structural members used to support live and dead loads. They are also defined as horizontal members that transfer floor loads to the support by flexure. It is also called a surface element which transfers the loads to the beams or column in case of flat slab. The type of slab depends on support condition, arrangement and load transfer mechanism etc. Slabs may be simply supported or continuous over one or more supports and are classified according to the method of supports as follows;

- ✓ Spanning one-way b/n beams or walls,
- ✓ Spanning two-way b/n the support beams or walls and
- ✓ Flat slabs carried on columns and edge beams or walls or with no interior beams.

There are two types of slabs based on the load transferring mechanisms.

- ➢ One Way
- ➢ Two Way Slabs.

One-way slabs transmit their load in one direction while two way slabs resist applied load in two directions. Mathematically,

- ➢ Ly/Lx >2 One Way Slab
- ➢ Ly/Lx <2 Two Way Slab

Where: **Lx**- shorter side

Ly- longer side

These types of slabs are composed of rectangular panels supported at all four edges by walls or beams stiff enough to be treated as unyielding. In this project paper most of the slabs are two way and need to be analyzed based on the principle of two way actions.

2.1 CLASSIFICATION OF SLAB

I. Solid slab
II. Ribbed slab
III. Pre casted slab
IV. Flat slab

Solid slab

These types of slabs are provided on the circulation core, which is estimated to be exposed to high live load. The reinforcements at the bottom cover all the area to form a mesh. Whereas the top (negative) reinforcement length used for continuous and end edge of the slab are not the same. For continuous slabs, one-third of the span length and for that of end edge slabs the bottom bars bent up and measured one-fourth of the span.

- In our project, we have only **solid slab.**

2.2 Slab analysis and Design

Step 1: **- Panel Layout**

This refers sketching the position plan from the given Architectural drawing. This helps us to know the location of columns, beams, No of panels and their support condition.

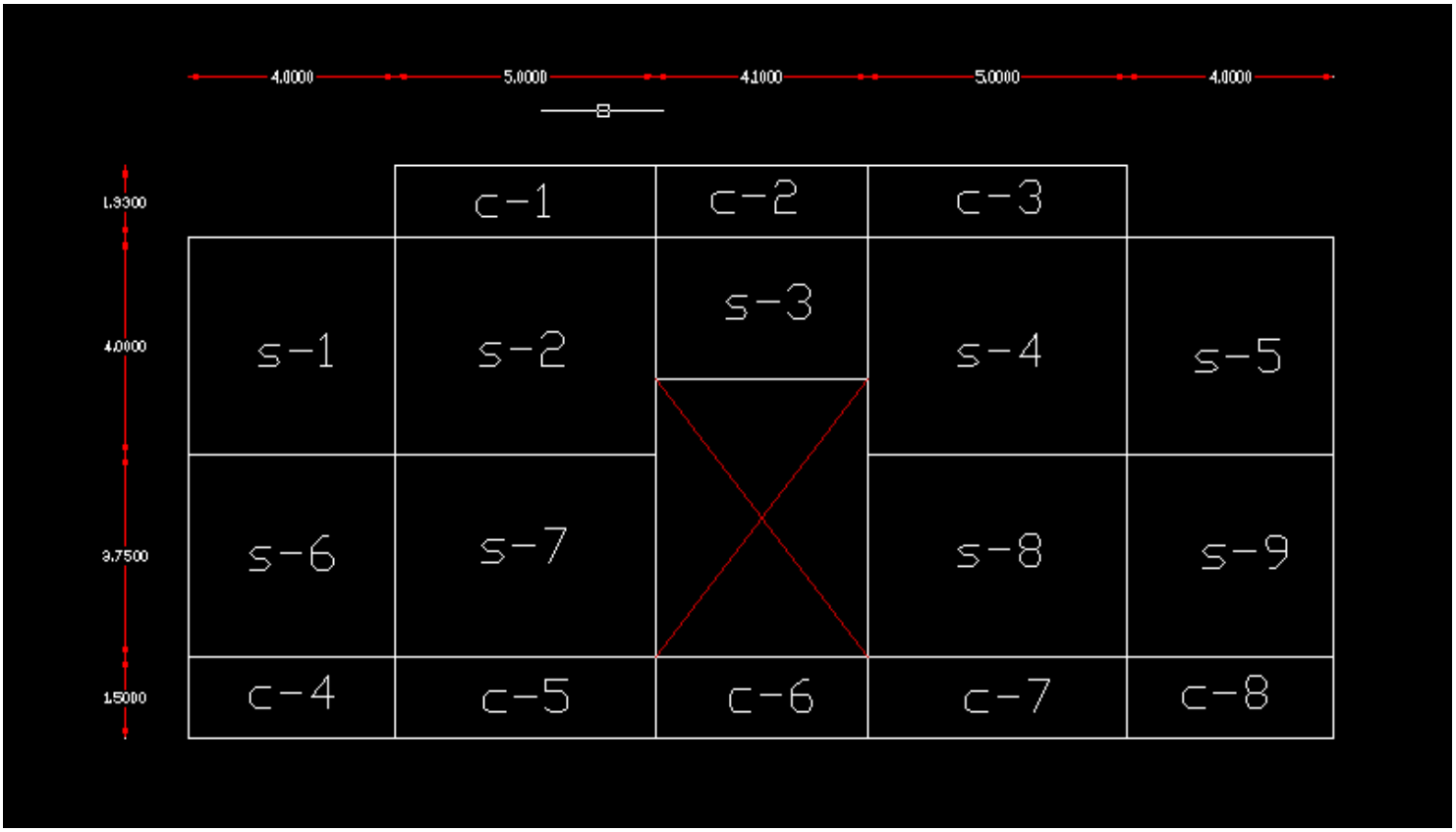
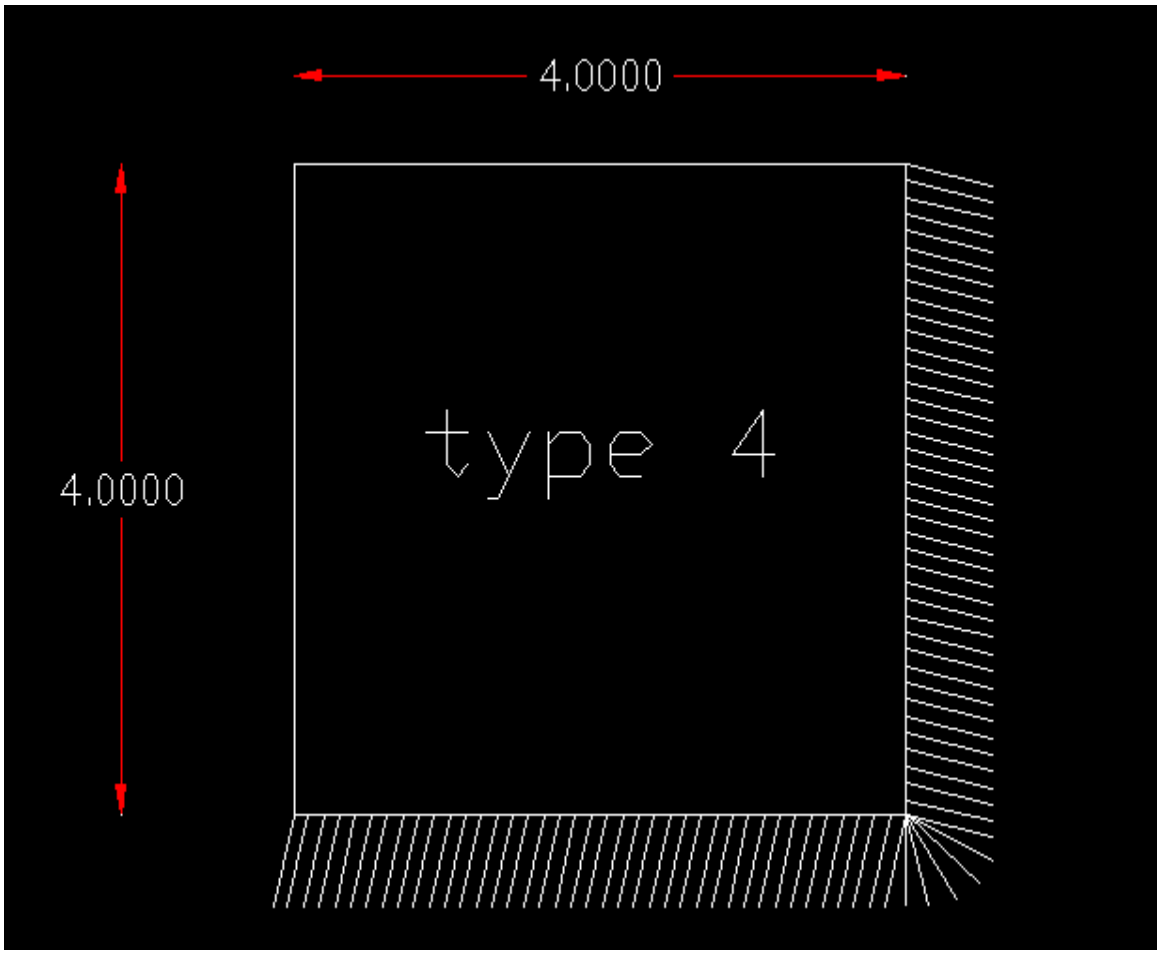


Figure 2.1 typical Floor Panel Layout

Step 2: - Check for the type of slab

- **We have sample calculation for S-1**



- **Ly=**4.00 - END
- Lx=4.00 - K=1.3
- Ratio=Ly/Lx - Cf=26

4/4=1 **<u>two-way</u>**

Step 2: - Check for d as per deflection requirement

- d=Lx/K*Cf

4000/1.3*26

d= <u>118.34</u> mm

Panel			Ly	Lx	Ly/Lx	Slab Type	Support Condition	K	C.F	Effective depth
	Type=	**4**	**4**	**4**	**1.00**	**Two Way**	**End**	**1.3**	**26**	**118.34**
	0									
0	S-1	1								
	1									
	Type=	2	5	4	1.25	Two Way	End	1.3	26	118.34
	1									
1	S-2	0								
	1									
	Type=	3	4.5	2.6	1.73	Two Way	End	1.3	26	76.92
	1									
1	S-3	1								
	0									
	Type=	2	5	4	1.25	Two Way	End	1.3	26	118.34
	1									
0	S-4	1								
	1									
	Type=	4	4	4	1.00	Two Way	End	1.3	26	118.34
	0									
1	S-5	0								
	1									
	Type=	2	4	3.75	1.07	Two Way	End	1.3	26	110.95
	1									
0	S-6	1								
	1									
	Type=	2	5	3.75	1.33	Two Way	End	1.3	26	110.95
	1									
1	S-7	0								
	1									
	Type=	2	5	3.75	1.33	Two Way	End	1.3	26	110.95
	1									
0	S-8	1								
	1									
	Type=	2	4	3.75	1.07	Two Way	End	1.3	26	110.95
	1									
1	S-9	0								
	1									

	Type=	C	5	1.33	3.76	Catilever	End	0.4	14	237.50
	0									
0	C-1	0								
	1									
	Type=	C	4.1	1.33	3.08	Catilever	End	0.4	14	237.50
	0									
1	C-2	1								
	1									
	Type=	C	5	1.33	3.76	Catilever	End	0.4	14	237.50
	0									
1	C-3	0								
	1									
	Type=	C	4	1.5	2.67	Catilever	End	0.4	14	267.86
	1									
0	C-4	0								
	0									
	Type=	C	5	1.5	3.33	Catilever	End	0.4	14	267.86
	1									
1	C-5	1								
	0									
	Type=	C	4.1	1.5	2.73	Catilever	End	0.4	14	267.86
	0									
0	C-6	1								
	1									
	Type=	C	4.5	1.5	3.00	Catilever	End	0.4	14	267.86
	1									
1	C-7	1								
	0									
	Type=	C	4	1.5	2.67	Catilever	End	0.4	14	267.86
	1									
1	C-8	0								
	0									

Concrete cover Determination

The nominal concrete cover is the distance b/n the surface of reinforcement closest to the nearest concrete surface = Cnom=C min + ΔCdev ESEN 1992: 2015 4.4.12(1)

Where; C min- minimum cover and

ΔC dev is allowable in design for deviation
Minimum cover, shall provide in order to ensure
Safe transmission of bond force

Corrosion resistance/ Durability

Fire resistance

$$C\ min = max\begin{cases} Cmin,b \\ Cmin,dur + \Delta Cdur,r - \Delta Cdur,st - \Delta Cdur,add \\ 10mm \end{cases}$$

Where;

C min, b -minimum cover due to bond requirement, see ES EN Art. 4.4.1.2 (3).

C min, dur - minimum cover due to environmental conditions, see ES EN Art 4.4.1.2 (5)
Δ C dur, γ- additive safety element, see ES EN Art 4.4.1.2 (6)
Δ C dur, st -reduction of minimum cover for use of stainless steel, see ES EN Art 4.4.1.2 (7)
ΔC dur, add -reduction of minimum cover for use of additional protection, see ES EN Art 4.4.1.2 (8)

But; the recommended value of ΔCdur, γ, ΔCdur, st, and ΔCdur, add is zero see Art. 4.4.1.2 (6, 7, and 8).

Cover Design for Bond

In order to transmit bond forces safely and to ensure adequate compaction of the concrete, the minimum cover should not be less than C min, b given in Table 4.2.EBCS, 2015. For separately placed bars minimum cover C min, b is equal to diameter of bar. Assuming 10mm diameter reinforcement bar, C min, b equals to 10mm.

Cover for durability

The value of minimum cover required for durability of reinforcement steel is determined using ES EN 1992:2015 table 4.4N. For structural class S3 and exposure class XC1 according to 4.1 of ES EN, C min, dur is 10mm.

Therefore, C min=**10mm**

Then Cmin=max$\begin{cases} Cmin, b = 10mm \\ Cmin, dur = 10mm \\ 10mm \end{cases}$

ΔCdev (allowance in Design for Variation)

From ES EN 2015 section 4.4.1.3 the recommended value is 10 mm.

Then; Cnom= Cmin+ ΔCdev

10mm+10mm=**20mm**

Cover Design for Fire

For the slab to sustain fire incident for 60 minutes the required cover and minimum height of the section can be determined from Table 5.8 of EBCS, 201.

For our structure capacity of fire resistance for REI=60 minute enough then

One - way slab C fire = 20mm

Two - way slab C fire=10mm&15mm

Slab thickness (hs) = 80mm

C fire = 20mm

Max d= 267.86 mm panel C5-C8

Use Cover=20 mm

Take ϕ 10 mm reinforcement

D=267.86+20+10/2=292.86=**300mm**

Note: The effective depth for deflection requirement is governed by the maximum of all, which is d=D-d' mm.

For all panel Over all depth (D) =**150mm** for safety against any unforeseen and For the purpose of construction simplicity and monolithic construction considerations, Over all depth of **150mm** is taken. d=150-20-5=**125mm**

❖ Cantilever section in the calculation part has an overall depth of 300mm.it is very large compare to panel slab depth, this variation came because of in EBSC 1992-2015 k value is smaller than other. We use for cantilever section same to slab but in check depth we came back not ok came change the depth.

2.3 Design load

LOAD CALCULATION

The slab is loaded with both dead load and live load. The dead load comes from the slab self-weight, floor finish, cement screed, plastering and partition wall. The live load coms from the function on panel so we will calculate live and dead load on each panel.

A. DEAD LOAD

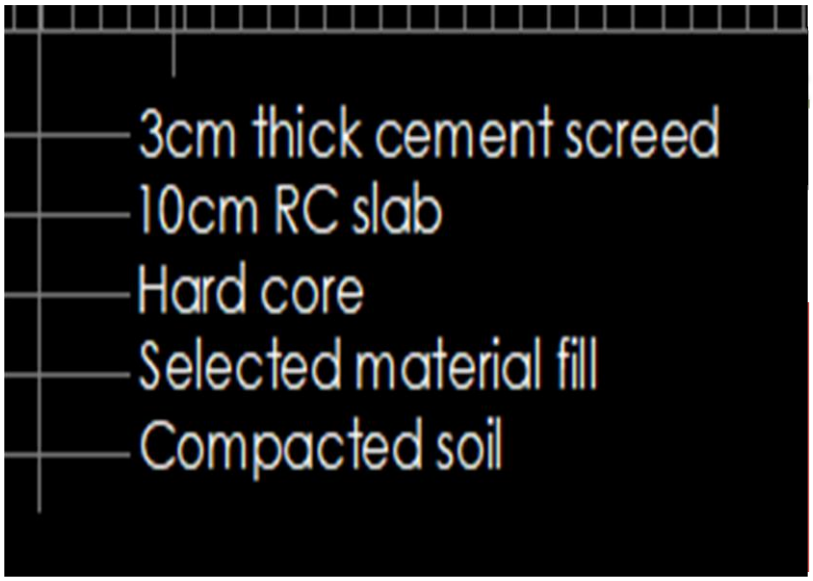


Figure 2.3 Slab section

Table 2.2 Categories of use from EBCS EN 1991-1-1:2014 table 6.1 page 12

Category	Specific Use	Example
A	Areas for domestic and residential activities	Rooms in residential buildings and houses; bedrooms and wards in hospitals; bedrooms in hotels and hostels kitchens and toilets.
B	Office areas	
C	Areas where people may congregate (with the exception of areas defined under category A, B, and D1))	**C1**: Areas with tables, etc. e.g. areas in schools, cafés, restaurants, dining halls, reading rooms, receptions. **C2**: Areas with fixed seats, e.g. areas in churches, theatres or cinemas, conference rooms, lecture halls, assembly halls, waiting rooms, railway waiting rooms. **C3:** Areas without obstacles for moving people, e.g. areas in museums, exhibition rooms, etc. and access areas in public and administration buildings, hotels, hospitals, railway station forecourts. **C4**: Areas with possible physical activities, e.g. dance halls, gymnastic rooms, stages. **C5**: Areas susceptible to large crowds, e.g. in buildings for public events like concert halls, sports halls including stands, terraces and access areas and railway platforms.
D	Shopping areas	**D1:** Areas in general retail shops **D2**: Areas in department stores
1) Attention is drawn to 6.3.1.1(2), in particular for C4 and C5. See EN 1990 when dynamic effects need to be considered. For Category E, see Table 6.3		
NOTE 1 Depending on their anticipated uses, areas likely to be categorized as C2, C3, C4 may be categorized as C5 by decision of the client and/or National annex. **NOTE 2** The National annex may provide sub categories to A, B, C1 to C5, D1 and D2 NOTE 3 See 6.3.2 for storage or industrial activity		

Table-2.3 Imposed loads on floors, balconies and stairs in buildings from EBCS EN 1991-1-1:2015 table 6.2 pages 21

Categories of loaded areas	***Qk* [kN/m2]**	***Qk* [kN]**
Category A **- Floors** **- Stairs** **- Balconies**	**1,5 to2,0** **2,0 to4,0** **2,5 to 4,0**	**2,0 to 3,0** **2,0 to 4,0** **2,0 to 3,0**
Category B	2,0 to 3,0	1,5 to 4,5
Category C - C1 - C2 - C3 - C4 - C5	2,0 to 3,0 3,0 to 4,0 3,0 to 5,0 4,5 to 5,0 5,0 to 7,5	3,0 to 4,0 2,5 to 7,0 (4,0) 4,0 to 7,0 3,5 to 7,0 3,5 to 4,5
category D - D1 - D2	4,0 to 5,0 4,0 to 5,0	3,5 to 7,0 (4,0) 3,5 to 7,0

Table-2.4 Live load in our building

Function	Bed room	LR & toilet	BR & LR	Corridor	Store	Living	Balcony	Kitchen	Toilet
Live Load kN/m2)	**2**	**2**	**2**	**3**	**5**	**2**	**2**	**2**	**2**

Function	Live Load kN/m2)
Bed room	**2**
LR & toilet	**2**
BR & LR	**2**
Corridor	**3**
Store	**5**
Living	**2**
Balcony	**2**
Kitchen	**2**
Toilet	**2**

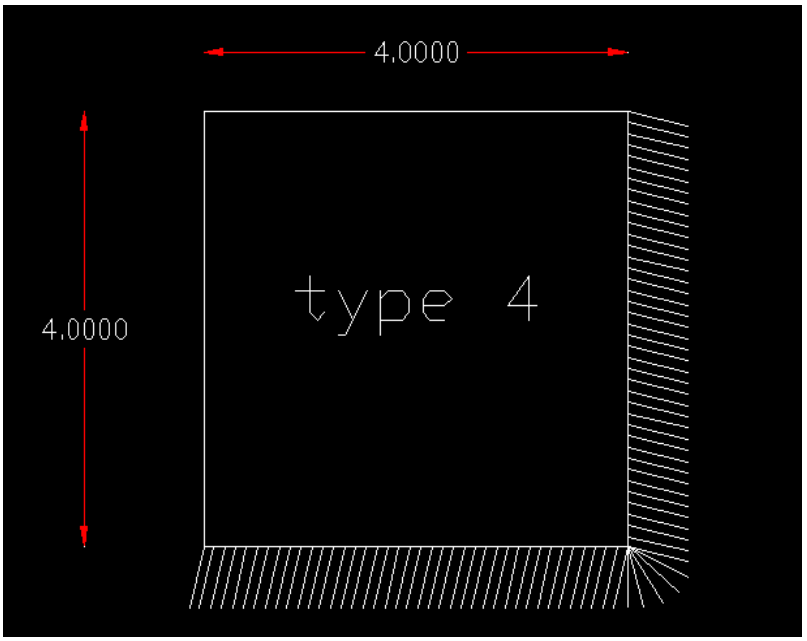
Table 2.5 **Dead load from EBCS EN 1991.1.1 2014**

Finishing	**unit wt(kN/m3)**	**Thicknes (mm)**	**Mortar Thic**
Clay Tile	**21**	**0.04**	**0.01**
Marble	**27**	**0.02**	**0.03**
Porcelain	**23**	**0.01**	**0.04**
Pvc	**16**	**0.002**	**0.048**
Rubber	**17**	**0.01**	**0.04**
Terrazo	**23**	**0.03**	**0.02**
Ceramic	**23**	**0.01**	**0.04**
Mortar	**23**		
RC	**25**		
HCB	**14**		
cement screed	**23**	**0.03**	**0.05**
Floor Height	**2.88**	**M**	

Design Load

- ✓ Self-Wight
- ✓ Partition load
- ✓ Live load

- **Design Load Computation for slab**



Dead load for **panel S-1**

- **Self-weight =D*γc**

  0.15*25=<u>3.75kN</u>/m2

- **Finishing floor mortal cement screed**

  =Thickness of mortal* γc + thickness of c. screed* γc.s

  =0.05*23+0.03*23

  =1.15+0.69

  =1.84 KN/m2

Plastering

=0.02*23

=0.46 KN/m2

- **Partition load**

=H(Tw*Lw* γc)+(Lw*Tp* γc)

Panel area

=2.88((0.1*2.99*14) + (3*0.04*23))

4*4

=2.88(4.186+2.76)

16

=1.25 KN/m2

- **Total dead load**

=the sum of the above

=(3.75+1.84+0.46+1.25) KN/m2

=7.3 KN/m^2

LIVE LOAD

Panel 1 is Toilet = **2 KN/m^2**

- Faunally Calculate Design Load

Pd=1.35DL+1.5LL

1.35*7.3 + 1.5*2

Pd=12.86 KN/m^2

Table-2.6 Design Load

Design Load typical Floor

PANEL	**Partition Load (point load)**	**FF Type**	**Se.wt+FF+Mo+Cei**	**Gk**	**Function**	**Qk**	**Design Load**
S-1	1.25	Cement screed	6.05	7.3	Bed Room & Living Room	2	**12.86**
S-2	1.9	Cement screed	6.05	7.95	Bed Room & Living Room	2	**13.73**
S-3	3.4	Cement screed	6.05	9.45	Kitchen	2	**15.75**
S-4	1.9	Cement screed	6.05	7.95	Bed Room & Living Room	2	**13.73**
S-5	1.25	Cement screed	6.05	7.3	Bed Room & Living Room	2	**12.86**
S-6	2.16	Cement screed	6.05	8.21	Living Room & Toilet	2	**14.08**
S-7	1.05	Cement screed	6.05	7.1	Living Room & Toilet	2	**12.59**
S-8	1.05	Cement screed	6.05	7.1	Living Room & Toilet	2	**12.59**
S-9	2.16	Cement screed	6.05	8.21	Living Room & Toilet	2	**14.08**
C-1	1.34	Cement screed	6.05	7.39	Bed Room	2	**12.97**
C-2	2.61	Cement screed	6.05	8.66	Kitchen	2	**14.7**
C-3	1.34	Cement screed	6.05	7.39	Bed Room	2	**12.97**
C-4	1.67	Cement screed	6.05	7.72	Kitchen	2	**13.42**
C-5	0	terrazzo	5.36	5.36	Corridor	3	**11.74**
C-6	0	terrazzo	5.36	5.36	Corridor	3	**11.74**
C-7	0	terrazzo	5.36	5.36	Corridor	3	**11.74**
C-8	1.67	Cement screed	6.05	7.72	Kitchen	2	**13.42**

Step 3: - Moment calculation

Moments for individual panels with edges either simply supported or fully fixed are calculated as:

$$Ms=\beta s*q*Lx^2$$

Where;

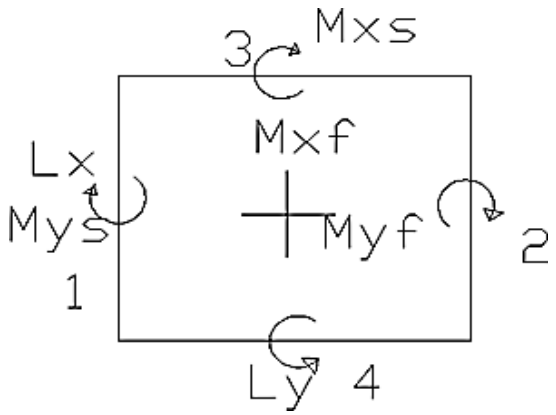
Ms- is the design moment per unit width at the point of reference

βs - is the coefficient given

q - is design load

Lx -is the shortest span of the pane

Ly -is the longer span of the panel



Therefore; $Mxf= \alpha_{xf}ql_x^2$ $Mxs= \alpha_{xs}ql_x^2$

$Myf= \alpha_{yf}ql_x^2$ $Mys= \alpha_{ys}ql_x^2$

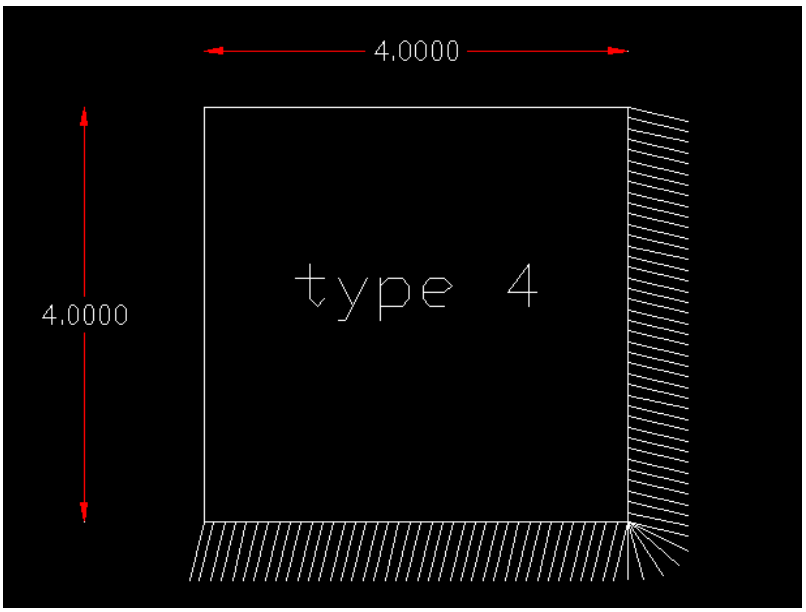
If the slab connected with an external wall or if any of its edge is partly fixed & partly simply supported, the following procedure may be adopted:

1. The ratio of the actual support moment to the bending moment of fully fixed slab, or the ratio of the width of fixed to the width of the simply supported part of the edge is evaluated.
2. The bending moments of the slab are then computed by interpolating between different conditions. [EBCS-2, 2015, Art 3.3.

Moment distribution for each panel

INDIVIDUAL Panel MOMENT

Sample calculation for individual pane moment, this panel taken from roof slab and it shows the step for calculating individual panel moment. The same procedure will be apply for other panels.



$Ms=\beta s*q*Lx^2$

For panel S-1, Ly/Lx=4/4=1, Pd=12.86KN/m^2

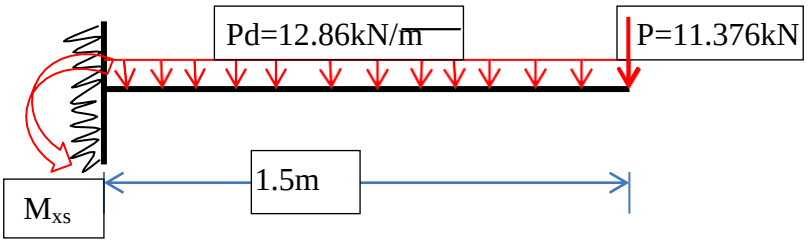
αxf=0.036, Mxf=0.036*12.86*4^2= 7.41**KNM**

αxs=0.047 , Mxf=0.047*12.86*4^2= 9.67**KNM**

αyf=0.034, Mxf=0.034*12.86*4^2= 7.0 **KN-M**

αys=0.045, Mxf=0.045*12.86*4^2= 9.26**KNM**

- For cantilever slabs from ground, floor C-4



$$Mxs = \frac{Pd\ Lx^2}{2} + PLx$$

$$= \frac{12.86 * 1.5^2}{2} + 11.376 * 1.5$$

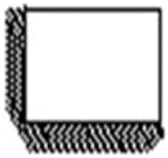
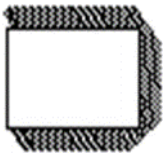
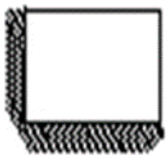
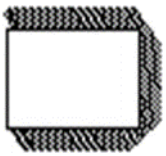
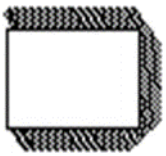
$$= 14.4675 + 17.2$$

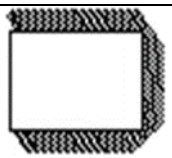
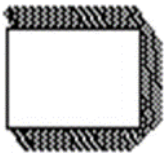
$$= 32.16$$

Individual Pane Moment Calculation.

Table 2.8 Individual panel Moment Calculation for Roof Slab

Individual Pane Moment Calculation For 1st Floor

S-1		1	12.86	αxf=	0.036	Mxf=	7.32
				αxs=	0.047	Mxs=	9.76
				αyf=	0.034	Myf=	7.00
				αys=	0.045	Mys=	9.33
S-2		1.25	13.73	αxf=	0.038	Mxf=	8.25
				αxs=	0.050	Mxs=	11.00
				αyf=	0.028	Myf=	6.04
				αys=	0.037	Mys=	8.05
S-3		1.73	15.75	αxf=	0.061	Mxf=	6.51
				αxs=	0.082	Mxs=	8.68
				αyf=	0.028	Myf=	2.93
				αys=	0.037	Mys=	3.90
S-4		1.25	13.73	αxf=	0.038	Mxf=	8.25
				αxs=	0.050	Mxs=	11.00
				αyf=	0.028	Myf=	6.04
				αys=	0.037	Mys=	8.05
S-5		1	12.86	αxf=	0.036	Mxf=	7.32
				αxs=	0.047	Mxs=	9.76
				αyf=	0.034	Myf=	7.00
				αys=	0.045	Mys=	9.33
S-6		1.07	14.08	αxf=	0.032	Mxf=	6.29
				αxs=	0.042	Mxs=	8.38
				αyf=	0.028	Myf=	5.45
				αys=	0.037	Mys=	7.26
S-7		1.33	12.59	αxf=	0.040	Mxf=	7.02
				αxs=	0.053	Mxs=	9.36
				αyf=	0.028	Myf=	4.87
				αys=	0.037	Mys=	6.49

S-8		1.33	12.59	αxf=	0.040	Mxf=	7.02
				αxs=	0.053	Mxs=	9.36
				αyf=	0.028	Myf=	4.87
				αys=	0.037	Mys=	6.49
S-9		1.07	14.08	αxf=	0.032	Mxf=	6.29
				αxs=	0.042	Mxs=	8.38
				αyf=	0.028	Myf=	5.45
				αys=	0.037	Mys=	7.26
C-1		3.76	12.97	αxs=	0.117	Mxs=	26.60
C-2		3.08	14.70	αxs=	0.107	Mxs=	28.13
C-3		3.76	12.97	αxs=	0.115	Mxs=	26.60
C-4		2.67	13.42	αxs=	0.108	Mxs=	32.16
C-5		3.33	11.74	αxs=	0.109	Mxs=	13.20
C-6		2.73	11.74	αxs=	0.106	Mxs=	13.20
C-7		3.00	11.74	αxs=	0.106	Mxs=	13.20
C-8		2.67	13.42	αxs=	0.105	Mxs=	32.16

MOMENT ADJUSTEMENT

For each support over which the slab is continuous there will thus generally be two different support moments. The difference may be distributed between the panels on either side of the support to equalize their moments, as in the moment distribution method for frames.

Two methods of differing accuracy are given here for treating the effects of this redistribution on moments away from the support. EBCS EN 2, 2015 article A.3.3.1 No.3 and 4

$\Delta M = (M_R - M_L)/M_R * 100$

Case 1: - If $\Delta M \leq 2$ use average moment then

$M_d = (M_R + M_L)/2$, $MR > M_L$.

Case 2: - If $\Delta M>20\%$ use moment distribution or distribute based on relative stiffness.

$M_d = M_R - K_R/(K_R+K_L) * \Delta M$… considering the right moment

$M_d = ML + K_L/(K_R+K_L) * \Delta M$... considering the left moment

Where $k=I/L_x$

If adjustment is b/n two way & cantilever, then $M_d=M_{max}$

Distribution Factor

$DF_R= (1/L_R) / (1/L_R+1/L_L)$

Sample moment adjustment between panel 1 and panel 2

$\Delta M=9.33-8.05$

$=1.28$ M large=9.33

M large*0.2

=9.33*0.2

=1.86

$\Delta M<$M large -------use average method

- Average (9.33+8.05)/2 =8.69

For slab 1 and 6

$\Delta M=9.76-8.38$

$=1.38$ M large=9.76

M large*0.2

=9.76*0.2

=1.952

$\Delta M<$M large -------use average method

- Average (9.76+8.38)/2= 9.07

Table-2.9 Distribution Factor

Panel	L_R	L_L	DF_R
S1-S2	5	4	0.444
S1-S6	4	3.75	0.48
S2-S3	4.1	5	0.4439
S2-S7	4	3.75	0.48
S3-S4	5	4.1	0.451
S4-S5	4	5	0.55
S4-S8	4	3.75	0.48
S5-S9	4	3.75	0.48

Floor	B/N PANEL	MR	ML	(MR-ML)/ MR*100 %	Method of adjustment	Adjusted moment
Typical FLOOR	b/n S1 & S2	9.33	8.05	128	Average method	8.69
	b/n S1 & S6	9.76	8.38	138	>>	9.03
	b/n S2 & S3	8.05	3.9	415	Moment redistribution	6.18
	b/n S2 & S7	11	9.36	164	Average method	10.18
	b/n S3 & S4	8.05	3.9	415	Moment redistribution	6.18
	b/n S4 & S5	9.33	8.05	128	Average method	8.69
	b/n S4 & S8	11	9.36	164	Moment redistribution	7.27
	b/n S5 & S9	9.76	8.38	138	Moment redistribution	10.18
	b/n S6 & S7	7.26	6.49	77	>>	6.98
	b/n S8 & S9	7.26	6.49	77	>>	6.98

SPAN MOMENT ADJUSTEMENT

According to EBCS 2, 2015 article A.3.3.3 Method II

1. In this method, consideration of the effects of changes of support moments is limited to the adjacent spans. Since no effects on neighboring support sections need to be considered; only a simple balancing operation is required at each edge and no iterative process is involved.

2. The procedure for applying method II, is as follows

Support and span moments, are first calculated for individual panels by assuming each panel fully loaded. This is done by using the coefficients given.

The unbalanced moment is distributed using the moment distribution method. The relative stiffness of each panel shall be taken proportional to its gross moment of inertia divided by the smaller span.

If the support moment is decreased, the span moments m_{xf} and m_{yf} are then increased to allow for the changes of support moments. This increase is calculated as being equal to the change of the support moment multiplied by the factors given. If a support moment is increased, no adjustments shall be made to the span moments.

The change in the respective moments will be described as follows:

$\triangle M_{xf} = C_x \triangle M$

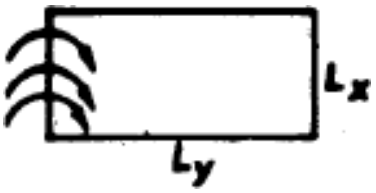
$\triangle M_{yf} = C_y \triangle M$

Adjusted field moment: $M_{xf} = M_{xf} + \triangle M_{xf}$

$\triangle M_{xf} = C_x \triangle M_{xs} + C_x \triangle M_{ys}$

$M_{yf} = M_{yf} + \triangle M_{yf}$

$\triangle M_{yf} = C_y \triangle M_{xs} + C_y \triangle M_{ys}$



For panel S1

Calculated support moments

Mxs= 9.76 Mys= 9.33

Adjusted support moments

Mxs =9.07 Mys =8.69

Ly/Lx= 1

Cx	Cy	Cx	Cy
0.380	**0.280**	**0.280**	**0.380**

$\triangle M_{xf} = C_x \triangle M_{xs} + C_x \triangle M_{ys}$

$=0.38(9.76-9.07)+(9.33-8.69)$

$=0.4414$

$M_{xf} = M_{xf} + \triangle M_{xf}$

$=7.32 + 0.4414$

$=7.7614$

$\triangle M_{yf} = C_y \triangle M_{xs} + C_y \triangle M_{ys}$

$=0.28(9.76-9.07)+0.38(9.33-8.69)$

$=0.1932+0.2432$

$=0.4364$

$M_{yf} = M_{yf} + \triangle M_{yf}$

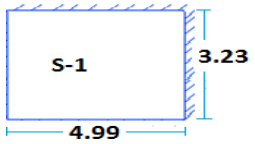
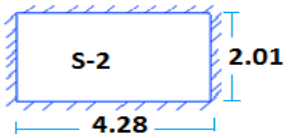
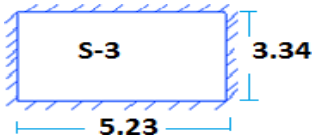
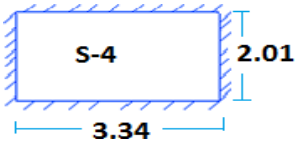
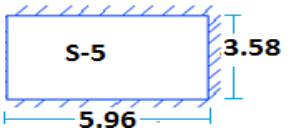
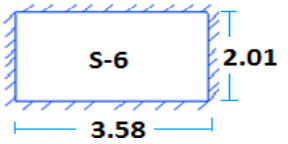
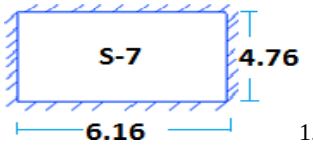
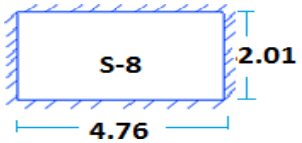
$=7.00+0.4364$

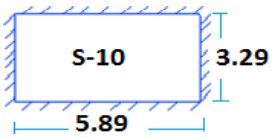
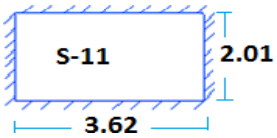
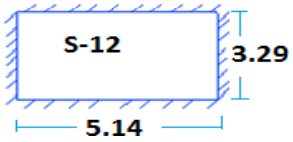
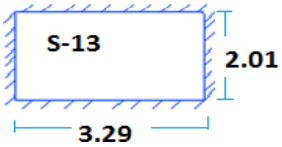
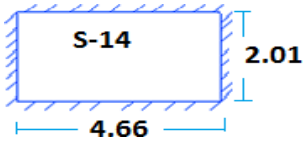
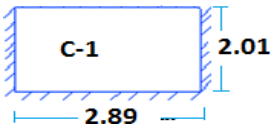
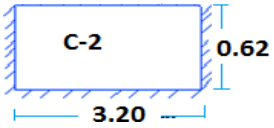
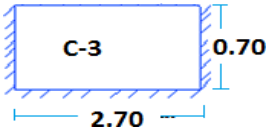
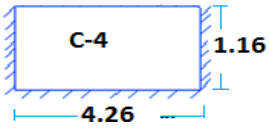
$=7.4364$

- Since the adjusted support moment exceeds the calculated moment no adjustment in the field moments is needed.

Panel	Ly/Lx	Cx	Cy	Cx	Cy
1	1.54	0.301	0.089	0.44	0.299
2	2.03	0	0	0	0
3	1.57	0.303	0.092	0	0
4	1.66	0	0	0.453	0.281
5	1.66	0.289	0.078	0	0
6	1.74	0.466	0.255	0	0
7	1.29	0.326	0.112	0	0
8	2.37	0	0	0	0
9	1.79	0	0	0	0
10	1.8	0	0	0	0
11	1.56	0.302	0.091	0	0
12	1.64	0	0	0	0
13	1.86	0.264	0.063	0	0
14	1.44	0	0	0	0

Table-2.10 Adjusted Moment Adjusted Moment roof floor

Panel	support condition	(Ly/Lx)	Pd (KN/m)	Adjusted Moment
S-1		1.00	12.86	**7.76**
				9.07
				7.44
				8.69
S-2		1.25	13.73	**9.14**
				26.60
				8.63
				8.69
S-3		1.73	15.75	**6.51**
				28.13
				2.93
				6.18
S-4		1.25	13.73	**9.14**
				26.60
				8.63
				6.18
S-5		1	12.86	**7.76**
				0.00
				7.44
				8.69
S-6		1.07	14.08	**6.29**
				9.07
				5.5.
				0.00
S-7	 1.33	1.33	12.59	**7.02**
				10.18
				4.87
				6.98
S-8		1.33	12.59	**7.02**
				7.27
				4.87
				0.00

S-9		1.07	14.08	**6.29** **10.18** **5.53** **6.98**
C-1		3.76	12.97	**26.6**
C-2		3.08	14.70	**28.13**
C-3		3.76	12.97	**26.6**
C-4		2.67	13.42	**32.16**
C-5		3.33	11.74	**13.42**
C-6		2.73	11.74	**13.42**
C-7		3	11.74	**13.42**
C-8		2.67	13.42	**32.16**

Check depth for flexure

d= 150-20-5= 125

$d_{min} = \sqrt{Mmax/(0.295 * b * Fcd)}$

$=\sqrt{32160000/(0.295 * 1000 * 11.33)}$

$=\sqrt{32160000/(3342.35)}$

$=\sqrt{9621.97}$

=98.09

- 98.09 < 125 ……. safe

Reinforcement calculation

Effective Depth

Minimum & Maximum Reinforcement Area

$As_{in} = 0.13\% bd = {}^{0.13}//$

$As_{in} = 0.26({}^{fct}/f)\ b, but$

$fct = 0.3(f)^{2/3} = 0.3(20)^{1/3} = .\ so$

$As_{in} = 0.26({}^{2.21}//$

$As_{ax} = 0.04A_c/$

- $K_m = \sqrt{{}^{M}/_{(d/1000)^2}}$
- Determine $_s$ from the Km-Ks Table
- Calculate the area of tension steel required from $\boldsymbol{As = \frac{M * K_s}{d/1000}}$
- Select suitable sizes
- Spacing

$$S = \frac{b * as}{As}, where\ as = {}^{\pi d^2}/_4$$

- Provide area =412.56

Table-2.9 Reinforcement calculation

1ST FLOOR

Panel	Adjusted Moment	Φ	D	km	ks	As required	Spacing	As provided
S-1	**7.76**	8	125	22.11	2.975	183.26	Φ8 c/c 250	186.17
	9.07	10	125	24.09	2.985	216.59	Φ10 c/c 250	224.40
	7.44	8	118	23.11	2.980	187.80	Φ8 c/c 250	193.33
	8.69	10	115	25.63	2.990	225.94	Φ10 c/c 250	231.00
S-2	**9.14**	8	125	23.99	2.980	216.16	Φ8 c/c 230	218.55
	26.60	10	125	41.26	3.145	669.26	Φ10 c/c 110	714.00
	8.63	8	118	24.90	2.985	218.30	Φ8 c/c 230	218.55
	8.69	10	115	25.63	2.990	225.94	Φ10 c/c 250	231.00
S-3	**6.51**	8	125	20.25	2.965	181.03	Φ8 c/c 250	186.17
	28.13	10	125	42.43	3.160	711.13	Φ10 c/c 110	714.00
	2.93	8	118	14.50	2.960	169.54	Φ8 c/c 250	173.33
	6.18	10	115	21.62	2.970	165.23	Φ10 c/c 250	224.40
S-4	**9.14**	8	125	23.99	2.980	216.16	Φ8 c/c 230	218.55
	26.60	10	125	41.26	3.145	669.26	Φ10 c/c 110	714.00
	8.63	8	118	24.90	2.985	218.30	Φ8 c/c 230	218.55
	6.18	10	115	21.62	2.970	165.23	Φ10 c/c 250	224.40
S-5	**7.76**	8	125	22.11	2.975	183.22	Φ8 c/c 250	186.17
	0.00	10	125	0.00	2.960	179.60	Φ10 c/c 250	224.40
	7.44	8	118	23.11	2.980	187.80	Φ8 c/c 250	193.33
	8.69	10	115	25.63	2.990	225.94	Φ10 c/c 250	231.00
S-6	**6.29**	8	125	19.90	2.960	181.03	Φ8 c/c 250	186.17
	9.07	10	125	24.09	2.985	216.59	Φ10 c/c 250	224.40
	5.5.	8	118	19.93	2.960	169.54	Φ8 c/c 250	173.33
	0.00	10	115	0.00	2.960	165.23	Φ10 c/c 250	224.40
	6.97	8	125	21.03	2.970	181.03	Φ8 c/c 250	186.17
	7.47	10	125	25.52	2.990	243.51	Φ10 c/c 250	245.44

S-7	**3.94**	8	118	18.70	2.960	169.54	Φ8 c/c 250	173.33
	8.02	10	115	22.97	2.975	180.57	Φ10 c/c 250	224.40
S-8	**7.02**	8	125	21.03	2.970	181.03	Φ8 c/c 250	186.17
	10.18	10	125	21.57	2.970	179.60	Φ10 c/c 250	224.40
	4.87	8	118	18.70	2.960	169.54	Φ8 c/c 250	173.33
	6.98	10	115	0.00	2.960	165.23	Φ10 c/c 250	224.40
S-9	**7.02**	8	125	19.90	2.960	181.03	Φ8 c/c 250	186.17
	7.27	10	125	25.52	2.990	243.51	Φ10 c/c 250	245.44
	4.87	8	118	19.93	2.960	169.54	Φ8 c/c 250	173.33
	0.00	10	115	22.97	2.975	180.57	Φ10 c/c 250	224.40
C-1	**26.6**	10	125	0.00	2.960	181.03	Φ8 c/c 250	186.17
			125	41.26	3.145	669.33	Φ10 c/c 110	714.00
			118	0.00	2.960	169.54	Φ8 c/c 250	173.33
			115	0.00	2.960	165.23	Φ10 c/c 250	224.40
C-2	**28.13**	10	125	0.00	2.960	181.03	Φ8 c/c 250	186.17
			125	42.43	3.160	711.06	Φ10 c/c 110	714.00
			118	0.00	2.960	169.54	Φ8 c/c 250	173.33
			115	0.00	2.960	165.23	Φ10 c/c 250	224.40
C-3	**26.6**	12	125	0.00	2.960	181.03	Φ8 c/c 250	186.17
			125	41.26	3.145	669.33	Φ10 c/c 110	714.00
			118	0.00	2.960	169.54	Φ8 c/c 250	173.33
			115	0.00	2.960	165.23	Φ10 c/c 250	224.40
C-4	**32.16**	12	125	0.00	2.960	181.03	Φ8 c/c 250	186.17
			124	45.74	3.210	832.64	Φ12 c/c 130	869.98
			118	0.00	2.960	169.54	Φ8 c/c 250	173.33
			115	0.00	2.960	165.23	Φ10 c/c 250	224.40
C-5	**13.2**	10	125	0.00	2.960	181.03	Φ8 c/c 250	186.17
			125	29.07	3.030	320.04	Φ10 c/c 240	327.25
			118	0.00	2.960	169.54	Φ8 c/c 250	173.33
			115	0.00	2.960	165.23	Φ10 c/c 250	224.40
			125	0.00	2.960	181.03	Φ8 c/c 250	186.17
			125	29.07	3.030	320.04	Φ10 c/c 240	327.25

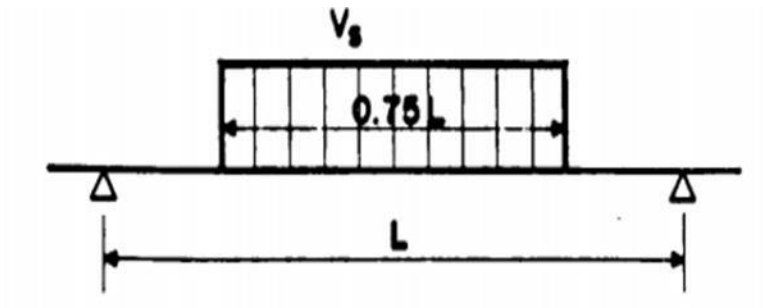
C-6	**13.2**	10	118	0.00	2.960	169.54	Φ8 c/c 250	173.33
			115	0.00	2.960	165.23	Φ10 c/c 250	224.40
C-7	**13.2**	10	125	0.00	2.960	181.03	Φ8 c/c 250	186.17
			125	29.07	3.030	320.04	Φ10 c/c 240	327.25
			118	0.00	2.960	169.54	Φ8 c/c 250	173.33
			115	0.00	2.960	165.23	Φ10 c/c 250	224.40
C-8	**32.16**	12	125	0.00	2.960	181.03	Φ8 c/c 250	186.17
			124	45.74	3.210	832.64	Φ12 c/c 130	869.98
			118	0.00	2.960	169.54	Φ8 c/c 250	173.33
			115	0.00	2.960	165.23	Φ10 c/c 250	224.40

Load transfer

The actual loads on the slab are transferred to beams as shear. The shear is calculated as:

$Vx = ß_{vx*} Pd*Lx$ $Vy = ß_{vy*} Pd*Ly$

The design load on a beam determined in the above may be taken as the maximum shear in the slab at the support, which will be distributed on 75% of the span of the beam.



Distribution of Load on Beam Supporting A Two-Way Spanning Slab

TYPICAL

FLOOR For

Panel S1

- ✓ Pd= 12.86 KN/m^2
- ✓ Ly= 4m
- ✓ Lx = 4m
- ✓ Ly/Lx = 1

βvx,c=0.400

βvx,d=0.267

βvy,c=0.400

βvy,d=0.26

Vx,c= βvx,c*Pd* Lx

Vx,c = 0.400*12.86*4= **20.576 KN/m**

Vx,d= βvx,d*Pd*Lx =0.267*12.86*4 =**13.73448KN/m**

Vy,c = βvy*Pd*Lx = 0.4*12.86*4 = **20.576KN/m**

Vy,d= βvy,d*Pd*Lx = 0.26*12.86*4 = **13.3744KN/m**

Table-2.12 Load transfer calculation from roof up to ground floor

1ST FLOOR								
Panel	**Ly**	**Lx**	**Ly/Lx**	**Design Load**			**Shear**	
S-1	4	4	1.23	**14.59**	βvx,c=	0.400	Vx,c=	20.57
					βvx,d=	0.267	Vx,d=	13.72
					βvy,c=	0.400	Vy,c=	20.57
					βvy,d=	0.260	Vy,d=	13.37
S-2	5	4	1.05	**13.93**	βvx,c=	0.444	Vx,c=	24.41
					βvx,d=	0.000	Vx,d=	0.00
					βvy,c=	0.360	Vy,c=	19.77
					βvy,d=	0.240	Vy,d=	13.18

S-3	4.5	2.6	2.06	**18.35**	βvx,c=	0.640	Vx,c=	26.21
					βvx,d=	0.348	Vx,d=	14.27
					βvy,c=	0.360	Vy,c=	14.75
					βvy,d=	0.000	Vy,d=	0.00
S-4	5	4	2.02	**13.93**	βvx,c=	0.444	Vx,c=	24.41
					βvx,d=	0.000	Vx,d=	0.00
					βvy,c=	0.360	Vy,c=	19.77
					βvy,d=	0.240	Vy,d=	13.18
S-5	4	4	1.25	**15.87**	βvx,c=	0.400	Vx,c=	20.57
					βvx,d=	0.267	Vx,d=	13.72
					βvy,c=	0.400	Vy,c=	20.57
					βvy,d=	0.260	Vy,d=	13.37
S-6	4	3.75	1.13	**13.93**	βvx,c=	0.406	Vx,c=	21.46
					βvx,d=	0.000	Vx,d=	0.00
					βvy,c=	0.360	Vy,c=	19.01
					βvy,d=	0.240	Vy,d=	12.68
S-7	5	3.75	1.59	**13.93**	βvx,c=	0.458	Vx,c=	21.63
					βvx,d=	0.000	Vx,d=	0.00
					βvy,c=	0.360	Vy,c=	16.99
					βvy,d=	0.240	Vy,d=	11.33
S-8	5	3.75	1.68	**21.31**	βvx,c=	0.458	Vx,c=	21.63
					βvx,d=	0.000	Vx,d=	0.00
					βvy,c=	0.360	Vy,c=	16.99
					βvy,d=	0.240	Vy,d=	11.33
S-9	4	3.75	2.03	**15.43**	βvx,c=	0.406	Vx,c=	21.46
					βvx,d=	0.000	Vx,d=	0.00
					βvy,c=	0.360	Vy,c=	19.01
					βvy,d=	0.240	Vy,d=	12.68

C-1	5	1.33	3.76	**12.97**	βvx,c=	FALSE	Vx,c=	28.63
					βvx,d=	FALSE	Vx,d=	0.00
					βvy,c=	FALSE	Vy,c=	0.00
					βvy,d=	FALSE	Vy,d=	0.00
C-2	4.1	1.33	3.08	**14.7**	βvx,c=	FALSE	Vx,c=	30.92
					βvx,d=	FALSE	Vx,d=	0.00
					βvy,c=	FALSE	Vy,c=	0.00
					βvy,d=	FALSE	Vy,d=	0.00
C-3	5	1.33	3.76	**12.97**	βvx,c=	FALSE	Vx,c=	28.63
					βvx,d=	FALSE	Vx,d=	0.00
					βvy,c=	FALSE	Vy,c=	0.00
					βvy,d=	FALSE	Vy,d=	0.00
C-4	4	1.55	2.67	**13.42**	βvx,c=	FALSE	Vx,c=	31.51
					βvx,d=	FALSE	Vx,d=	0.00
					βvy,c=	FALSE	Vy,c=	0.00
					βvy,d=	FALSE	Vy,d=	0.00
C-5	5	1.5	3.33	**11.74**	βvx,c=	FALSE	Vx,c=	17.60
					βvx,d=	FALSE	Vx,d=	0.00
					βvy,c=	FALSE	Vy,c=	0.00
					βvy,d=	FALSE	Vy,d=	0.00
C-6	4.1	1.5	2.73	**11.74**	βvx,c=	FALSE	Vx,c=	17.60
					βvx,d=	FALSE	Vx,d=	0.00
					βvy,c=	FALSE	Vy,c=	0.00
					βvy,d=	FALSE	Vy,d=	0.00
C-7	4.5	1.5	3	11.74	βvx,c=	FALSE	Vx,c=	17.60
					βvx,d=	FALSE	Vx,d=	0.00
					βvy,c=	FALSE	Vy,c=	0.00
					βvy,d=	FALSE	Vy,d=	0.00
C-8	4	1.5	2.67	**13.42**	βvx,c=	FALSE	Vx,c=	31.51
					βvx,d=	FALSE	Vx,d=	0.00
					βvy,c=	FALSE	Vy,c=	0.00
					βvy,d=	FALSE	Vy,d=	0.00

Shear Design Resistance

- $_{RD,}$ = (0.12 (100 $_1Fc$)$^{1/}$3) bd, but

= 1 + ($^{200}/d$)$^{0.5}$ = 1 + ($^{200}/125$)$^{0.5}$

= 2.265 > 2

TAKEV K=2

p_1=Max { 0.26 Fctm/Fyk=(0.26*2.21)/400=0.0014365 ; 0.0013 }

- $_{RD,}$ = (0.12* 2 (100 *0.0014365* 20)$^{1/}$3) *1* 125

= 43.39 $\boldsymbol{KN}$

- $_{MIN}$ = 0.035(k)$^{1.5}$ ((Fck)$^{1/}$2) bd

= 0.035 * 2$^{1.5}$ * 20$^{0.5}$ * 1 * 125

= 55.78$\boldsymbol{KN}$

$_{RD,}$ = 43.39kN < $_{ED}$ = 55.78 kN

Therefore, shear reinforcement is required.

<u>Deflection</u>

- L/d(required) =1.3(11+1.5(20)$^{0.5}$ *0.447/0.148 +3.2(20)$^{0.5}$ ((0.447/0.148)-1)

= 1.3(31.3 + 41.1)

=94.12 *1.25 …….1.25…corrected factor

119.845

- L/d(actual) = L_x *1000 / d

= 4*1000/126

31.74…….ok

Table-2.13 Shear Design Resistance calculation from roof up to ground floor

1ST FLOOR							
Panel	**Shear**		**d**	**As provided**	K	ρ1	Check
S-1	Vx,c=	20.57	126	186.17	1.300	0.148	SAFE!
	Vx,d=	13.72	125	224.40	1.300	0.180	SAFE!
	Vy,c=	20.57	118	193.33	1.300	0.164	SAFE!
	Vy,d=	13.37	115	231.00	1.300	0.201	SAFE!
S-2	Vx,c=	24.41	126	218.55	1.300	0.173	SAFE!
	Vx,d=	0.00	125	714.00	1.300	0.571	SAFE!
	Vy,c=	19.77	118	218.55	1.300	0.185	SAFE!
	Vy,d=	13.18	115	231.00	1.300	0.201	SAFE!
S-3	Vx,c=	26.21	126	186.17	1.300	0.148	SAFE!
	Vx,d=	14.27	125	714.00	1.300	0.571	SAFE!
	Vy,c=	14.75	118	173.33	1.300	0.147	SAFE!
	Vy,d=	0.00	115	224.40	1.300	0.195	SAFE!
S-4	Vx,c=	24.41	126	218.55	1.300	0.173	SAFE!
	Vx,d=	0.00	125	714.00	1.300	0.571	SAFE!
	Vy,c=	19.77	118	218.55	1.300	0.185	SAFE!
	Vy,d=	13.18	115	224.40	1.300	0.195	SAFE!
S-5	Vx,c=	20.57	126	186.17	1.300	0.148	SAFE!
	Vx,d=	13.72	125	224.40	1.300	0.180	SAFE!
	Vy,c=	20.57	118	193.33	1.300	0.164	SAFE!
	Vy,d=	13.37	115	231.00	1.300	0.201	SAFE!
S-6	Vx,c=	21.46	126	186.17	1.300	0.148	SAFE!
	Vx,d=	0.00	125	224.40	1.300	0.180	SAFE!
	Vy,c=	19.01	118	173.33	1.300	0.147	SAFE!
	Vy,d=	12.68	115	224.40	1.300	0.195	SAFE!
	Vx,c=	21.63	126	186.17	1.300	0.148	SAFE!
	Vx,d=	0.00	125	245.44	1.300	0.196	SAFE!

S-7	Vy,c=	16.99	118	173.33	1.300	0.147	SAFE!
	Vy,d=	11.33	115	224.40	1.300	0.195	SAFE!
S-8	Vx,c=	21.63	126	186.17	1.300	0.148	SAFE!
	Vx,d=	0.00	125	224.40	1.300	0.180	SAFE!
	Vy,c=	16.99	118	173.33	1.300	0.147	SAFE!
	Vy,d=	11.33	115	224.40	1.300	0.195	SAFE!
S-9	Vx,c=	21.46	126	186.17	1.300	0.148	SAFE!
	Vx,d=	0.00	125	245.44	1.300	0.196	SAFE!
	Vy,c=	19.01	118	173.33	1.300	0.147	SAFE!
	Vy,d=	12.68	115	224.40	1.300	0.195	SAFE!

C-1	Vx,c=	28.63	126	186.17	0.400	0.148	52.25	SAFE!
	Vx,d=	0.00	125	714.00	0.400	0.571	69.41	SAFE!
	Vy,c=	0.00	118	173.33	0.400	0.147	51.78	SAFE!
	Vy,d=	0.00	115	224.40	0.400	0.195	51.78	SAFE!
C-2	Vx,c=	30.92	126	186.17	0.400	0.148	51.78	SAFE!
	Vx,d=	0.00	125	714.00	0.400	0.571	66.14	SAFE!
	Vy,c=	0.00	118	173.33	0.400	0.147	51.78	SAFE!
	Vy,d=	0.00	115	224.40	0.400	0.195	51.78	SAFE!
C-3	Vx,c=	28.63	126	186.17	0.400	0.148	52.25	SAFE!
	Vx,d=	0.00	125	714.00	0.400	0.571	61.66	SAFE!
	Vy,c=	0.00	118	173.33	0.400	0.147	51.78	SAFE!
	Vy,d=	0.00	115	224.40	0.400	0.195	51.78	SAFE!
C-4	Vx,c=	31.51	126	186.17	0.400	0.148	52.25	SAFE!
	Vx,d=	0.00	125	869.98	0.400	0.702	52.25	SAFE!
	Vy,c=	0.00	118	173.33	0.400	0.147	51.78	SAFE!
	Vy,d=	0.00	115	224.40	0.400	0.195	51.78	SAFE!
C-5	Vx,c=	17.60	126	186.17	0.400	0.148	52.25	SAFE!
	Vx,d=	0.00	125	327.25	0.400	0.262	63.20	SAFE!
	Vy,c=	0.00	118	173.33	0.400	0.147	51.78	SAFE!
	Vy,d=	0.00	115	224.40	0.400	0.195	51.78	SAFE!
C-6	Vx,c=	17.60	126	186.17	0.400	0.148	52.25	SAFE!
	Vx,d=	0.00	124	327.25	0.400	0.262	52.25	SAFE!
	Vy,c=	0.00	118	173.33	0.400	0.147	51.78	SAFE!
	Vy,d=	0.00	115	224.40	0.400	0.195	51.78	SAFE!
C-7	Vx,c=	17.60	126	186.17	0.400	0.148	52.25	SAFE!
	Vx,d=	0.00	125	327.25	0.400	0.262	52.25	SAFE!
	Vy,c=	0.00	118	173.33	0.400	0.147	51.78	SAFE!
	Vy,d=	0.00	115	224.40	0.400	0.195	51.78	SAFE!
C-8	Vx,c=	31.51	126	186.17	0.400	0.148	52.25	SAFE!
	Vx,d=	0.00	125	869.98	0.400	0.702	69.41	SAFE!
	Vy,c=	0.00	118	173.33	0.400	0.147	51.78	SAFE!
	Vy,d=	0.00	115	224.40	0.400	0.195	51.78	SAFE!

Table-2.14 Deflection calculation from roof up to ground floor

Panel	Adjusted Moment	As required	As provided	ρ1	ρo	L/d (req)	L/d (Act)	Check
S-1	**7.76**	183.26	186.17	0.148	0.50	119.845	31.746	SAFE!
	9.07	216.59	224.40	0.180	0.50	90.526	32.000	SAFE!
	7.44	187.80	193.33	0.164	0.50	103.487	33.898	SAFE!
	8.69	225.94	231.00	0.201	0.50	75.379	34.783	SAFE!
S-2	**9.14**	216.16	218.55	0.173	0.50	93.112	31.746	SAFE!
	26.60	669.26	714.00	0.571	0.50	28.175	32.000	SAFE!
	8.63	218.30	218.55	0.185	0.50	83.417	42.373	SAFE!
	8.69	225.94	231.00	0.201	0.50	75.379	43.478	SAFE!
S-3	**6.51**	181.03	186.17	0.148	0.50	121.316	20.635	SAFE!
	28.13	711.13	714.00	0.571	0.50	26.516	20.800	SAFE!
	2.93	169.54	173.33	0.147	0.50	121.711	22.034	SAFE!
	6.18	165.23	224.40	0.195	0.50	92.406	22.609	SAFE!
S-4	**9.14**	216.16	218.55	0.173	0.50	93.112	31.746	SAFE!
	26.60	669.26	714.00	0.571	0.50	28.175	32.000	SAFE!
	8.63	218.30	218.55	0.185	0.50	83.417	42.373	SAFE!
	6.18	165.23	224.40	0.195	0.50	92.406	43.478	SAFE!
S-5	**7.76**	183.22	186.17	0.148	0.50	119.867	31.746	SAFE!
	0.00	179.60	224.40	0.180	0.50	104.852	32.000	SAFE!
	7.44	187.80	193.33	0.164	0.50	103.487	33.898	SAFE!
	8.69	225.94	231.00	0.201	0.50	75.379	34.783	SAFE!
S-6	**6.29**	181.03	186.17	0.148	0.50	121.316	29.762	SAFE!
	9.07	216.59	224.40	0.180	0.50	90.526	30.000	SAFE!
	5.5.	169.54	173.33	0.147	0.50	121.711	31.780	SAFE!
	0.00	165.23	224.40	0.195	0.50	92.406	32.609	SAFE!
S-7	**6.97**	181.03	186.17	0.148	0.50	121.316	29.762	SAFE!
	7.47	243.51	245.44	0.196	0.50	76.892	30.000	SAFE!
	3.94	169.54	173.33	0.147	0.50	121.711	42.373	SAFE!

	8.02	180.57	224.40	0.195	0.50	92.406	43.478	SAFE!

S-8	**7.02**	181.03	186.17	0.148	0.50	121.316	29.762	SAFE!
	10.18	179.60	224.40	0.180	0.50	104.852	30.000	SAFE!
	4.87	169.54	173.33	0.147	0.50	121.711	42.373	SAFE!
	6.98	165.23	224.40	0.195	0.50	92.406	43.478	SAFE!
S-9	**7.02**	181.03	186.17	0.148	0.50	121.316	29.762	SAFE!
	7.27	243.51	245.44	0.196	0.50	76.892	30.000	SAFE!
	4.87	169.54	173.33	0.147	0.50	121.711	33.898	SAFE!
	0.00	180.57	224.40	0.195	0.50	92.406	34.783	SAFE!
C-1	**26.6**	181.03	186.17	0.148	0.50	37.328	10.556	SAFE!
		669.33	714.00	0.571	0.50	8.668	10.640	SAFE!
		169.54	173.33	0.147	0.50	37.449	11.271	SAFE!
		165.23	224.40	0.195	0.50	28.433	11.565	SAFE!
C-2	**28.13**	181.03	186.17	0.148	0.50	37.328	10.556	SAFE!
		711.06	714.00	0.571	0.50	8.160	10.640	FALE!
		169.54	173.33	0.147	0.50	37.449	11.271	SAFE!
		165.23	224.40	0.195	0.50	28.433	11.565	SAFE!
C-3	**26.6**	181.03	186.17	0.148	0.50	37.328	10.556	SAFE!
		669.33	714.00	0.571	0.50	8.668	10.640	SAFE!
		169.54	173.33	0.147	0.50	37.449	11.271	SAFE!
		165.23	224.40	0.195	0.50	28.433	11.565	SAFE!
C-4	**32.16**	181.03	186.17	0.148	0.50	37.328	11.905	SAFE!
		832.64	869.98	0.702	0.50	7.980	12.097	SAFE!
		169.54	173.33	0.147	0.50	37.449	12.712	SAFE!
		165.23	224.40	0.195	0.50	28.433	13.043	SAFE!
C-5	**13.2**	181.03	186.17	0.148	0.50	37.328	11.905	SAFE!
		320.04	327.25	0.262	0.50	15.843	12.000	SAFE!
		169.54	173.33	0.147	0.50	37.449	12.712	SAFE!
		165.23	224.40	0.195	0.50	28.433	13.043	SAFE!
		181.03	186.17	0.148	0.50	37.328	11.905	SAFE!

C-6	**13.2**	320.04	327.25	0.262	0.50	15.843	12.000	SAFE!
		169.54	173.33	0.147	0.50	37.449	12.712	SAFE!
		165.23	224.40	0.195	0.50	28.433	13.043	SAFE!
C-7	**13.2**	181.03	186.17	0.148	0.50	37.328	11.905	SAFE!
		320.04	327.25	0.262	0.50	15.843	12.000	SAFE!
		169.54	173.33	0.147	0.50	37.449	12.712	SAFE!
		165.23	224.40	0.195	0.50	28.433	13.043	SAFE!
C-8	**32.16**	181.03	186.17	0.148	0.50	37.328	11.905	SAFE!
		832.64	869.98	0.702	0.50	7.980	12.097	SAFE!
		169.54	173.33	0.147	0.50	37.449	12.712	SAFE!
		165.23	224.40	0.195	0.50	28.433	13.043	SAFE!

Chapter Three

Design of Stair

Stair case is steps arranged in series and is an inclined structural system leading from one floor to another providing communication between floors. They should be provided with natural light and proper ventilation. Their location should be accessible for public buildings and centrally for easy access and privacy in residential buildings.

Common Terms

Tread

Going

Riser

Rise

Landing

Flight

The available data we have are

Thickness of the marble = 20mm

Thickness of plastering = 20mm

Thickness of cement screed = 30mm

Riser = 10*16cm=160cm

Tread = 10*30cm=300mm

Live load = 2kN/m2

Material: C-20/25, S-400

Depth determination

$\frac{lx}{d}$ = k [11+1.5$\sqrt{fck}\,\frac{\rho^{\circ}}{\rho-\rho'}$+$\frac{1}{12}\sqrt{fck}\sqrt{\left(\frac{\rho'}{\rho^{\circ}}\right)}$] *F1; $\rho < \rho^{\circ}$

$\frac{lx}{d}$ = k [11+1.5$\sqrt{fck}\,\frac{\rho^{\circ}}{\rho calc}$+3.2$\sqrt{fck}(\frac{\rho^{\circ}}{\rho calc})^{1.5}$] *F1; $\rho > \rho^{\circ}$

According to EC-2 table 7.4N take the value of K=1.0, concrete lightly stressed ρ=0.5% is 20 for simple supported slab of structural system.

Stair depth determination for inclined slab

Assume the slab is lightly reinforced (ρ=0.5%)

$$\rho_0 = \sqrt{f_{ck}} * 10^{-3}$$

$$= \sqrt{20} * 10^{-3}$$

$$= 4.47 * 10^{-3}$$

Since $\rho > \rho_0$ use

$$\frac{l}{d} = k[11+1.5\sqrt{fck}\,\frac{\rho_0}{\rho-\rho'} + \frac{1}{12}\quad fck\,(\sqrt{\frac{\rho'}{\rho_0}})]\ ,\ \rho > \rho_0$$

$\frac{lx}{d}$ = 1 [11+1.5$\sqrt{20}\,\frac{0.447}{0.5-0}$+$\frac{1}{12}\sqrt{20}\sqrt{\left(\frac{0}{0.447}\right)}$] *1.25 = 21.24

d= Lx/ ($\frac{lx}{d}$)=1200/21.24=56.497mm

Depth determination for landing slab

$\frac{lx}{d}$ = 1 [11+1.5$\sqrt{20}\,\frac{0.447}{0.5-0}$+$\frac{1}{12}\sqrt{20}\sqrt{\left(\frac{0}{0.447}\right)}$] *1.25 = 21.24

d= Lx/ ($\frac{lx}{d}$)=1400/21.24=65.9mm

depth determination for landing on slab

$\frac{lx}{d}$ = 1 [11+1.5$\sqrt{20}\,\frac{0.447}{0.5-0}$+$\frac{1}{12}\sqrt{20}\sqrt{\left(\frac{0}{0.447}\right)}$] *1.25 = 21.24

d= Lx/ ($\frac{lx}{d}$)=1270/21.24=59.79mm

depth of stair = max{56.497;65.9;59.79} =65.9mm

depth=70mm

load on inclined slab

Dead load

Cement screed

Thickness 55mm

γ = 23 KN/m^3

load=23 KN/m^3*0.055m=1.265 KN/m^2

on riser and thread = (0.15+0.3)*1.265*9=5.121KN/m

Floor finish load

Material type marble tiling

$\gamma = 27KN/m^3$

thickness 5mm

load $=0.135KN/m^3$

on riser and thread = (0.15+0.3)*0.135*9=0.5454KN/m

Self-weight

$\gamma = 25\ KN/m^3$

Depth=d + cover + Ø/2

=70+25+8/2=99mm

The minimum depth of a slab should be greater than120mm , and for construction simplicity we take the governing depth of the floor for the slab.

D=200mm

Load=D* γ=0.2m*25 KN/m^3=5 KN/m^2

Self-weight of Riser and thread

Load = (1/2) *0.3*0.15*25=0.5625KN/m

Total load = 9*0.5625=5.0625KN/m

Plastering

Thickness 2cm

$\gamma = 23\ KN/m^3$

load = 0.02*23=0.46 KN/m^2

Total dead load

=5.0625+0.5454+5.121=10.7314KN/m

Distributed load on inclined part

=10.7314KN/2.7m=3.975 KN/m^2

DL=3.975+5+0.46=9.435KN/m^2

Live load for residential building stair case is LL= 4 KN/m^2

Design load for inclined slab is Pd=1.35*DL+1.5*LL

=1.35*9.435+1.5*4

= 18.74 KN/m^2

Load calculation for landing slabs (right and left slabs)

Cement screed

Thickness 55mm

$\gamma = 23\ KN/m^3$

load=23 KN/m^3*0.055m=1.265 KN/m^2

Floor finish load

Material type marble tiling

$\gamma = 27\ KN/m^3$

thickness 5mm

load $=0.135\ KN/m^2$

Self-weight

$\gamma = 25\ KN/m^3$

Depth=d + cover + Ø/2

=70+25+8/2=99mm

The minimum depth of a slab should be greater than120mm, and for construction simplicity we take the governing depth of the floor for the slab.

D=200mm

Load=D* γ=0.2m*25KN/m^3=5 KN/m^2

Plastering

Thickness 2cm

γ = 23KN/m2

load = 0.02*23=0.46 KN/m^2

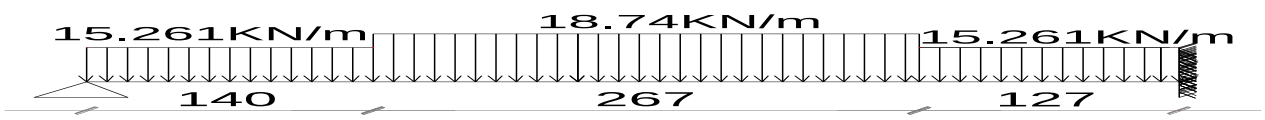
Total dead load

=5+0.46+0.135+1.265=6.86

Design load =1.35*DL+1.5*LL

=1.35*6.86+1.5*4

=15.261KN/m^2



Design For Flexure

Use Ø 14

D=200mm

d=200-25-14/2=168mm

b=1000mm

$a_s = \pi \emptyset^2/4$

$= 3.14*14^2/4 = 153.86mm^2$

C20/25

fcd=0.85*fyk/1.5

=0.85*20/1.5=11.33Mpa

fctm = 2.2Mpa

S400

fyd=400/1.15=347.83Mpa

+ve span moment

Msd=32.36Mpa

$\mu_{sd} = \frac{Msd}{(fcd*b*d*d)} = \frac{32.36KNm}{(11.33*1000mm*168*168mm)} = 0.102$

$Z = 0.5*167*(1+\sqrt{(1-2*0.102)}) = 159$

$As\ min = \frac{(0.26*2.2*1000*167)}{400} = 240.24$

$Ascal = \frac{32.36KNm}{159*347.83} = 585$ Asriq=max{240.24,585 } Asriq=585

$Scale = \frac{1000*\pi}{4*585} * 14^2 = 263$

Sprov=min { 3*D=200*3=600 ; 400 ; 263 } Sprov=263≈260

$Astuesd = 1000*\pi*14^2/(4*260) = 591.8$

$F1 = \frac{500}{400+(\frac{585}{591.8})} = 1.264$

$\rho_{calc\ (\%)} = (\frac{591.8}{1000*168})*100 = 0.352$

For $\rho_{calc} < \rho^{\circ}$, 0.28<0.5

$\frac{lx}{d} = 1.3\ [11+1.5\sqrt{20}\frac{0.5}{0.352-0}+\frac{1}{12}\sqrt{20}\sqrt{(\frac{0}{0.5})}]*1.264 = 40.4$

$dcalc = lx/\frac{lx}{d} = *1000/40.4 = 39.5$

In our case dused =171 > driq=39.5 so our depth more safe

<u>Use Ø14 clc 260</u>

<u>-ve support moment</u>

Msd=63.14Mpa

$\mu_{sd} = \frac{Msd}{(fcd*b*d*d)} = \frac{63.14KNm}{(11.33*1000mm*168*168mm)} = 0.1974$

$Z = 0.5*167*(1+\sqrt{(1-2*0.1974)}) = 149.3$

Asmin=$\frac{(0.26*2.2*1000*168)}{400}$=240.24

Ascal=$\frac{63.014Knm}{149.3*347.83}$=1215 Asriq=max{240.24,1215 } Asriq=1215

Scalc=$\frac{1000*\pi}{4*1215}$*14^2 =126.6

Sprov=min $\begin{cases} 3*D=200*3=600 \\ 400 \\ 126.6 \end{cases}$ Sprov=126.6≈120

Astuesd=1000*π*14^2/ (4*120) =591.8

F1=$\frac{500}{400+(\frac{1215}{1282.2})}$=1.3186

$\rho_{calc\ (\%)}$ = ($\frac{1252.2}{1000*168}$)*100=0.765

For $\rho_{calc} > \rho^{\circ}$, 0.28>0.5

$\frac{lx}{d}$ = 1.3 [11+1.5$\sqrt{20}\frac{0.5}{0.352-0}$+$\frac{1}{12}\sqrt{20}\sqrt{(\frac{0}{0.5})}$] *1.264=26.4

dcalc=lx/$\frac{lx}{d}$=*1000/26.4=39.5

In our case dused =171 > driq=39.5 so our depth more safe

<u>Use Ø14 clc 160</u>

Chapter Four

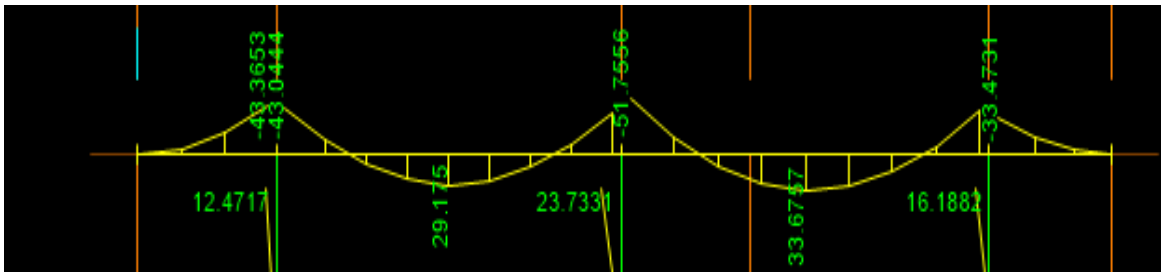
Beam and Column Design

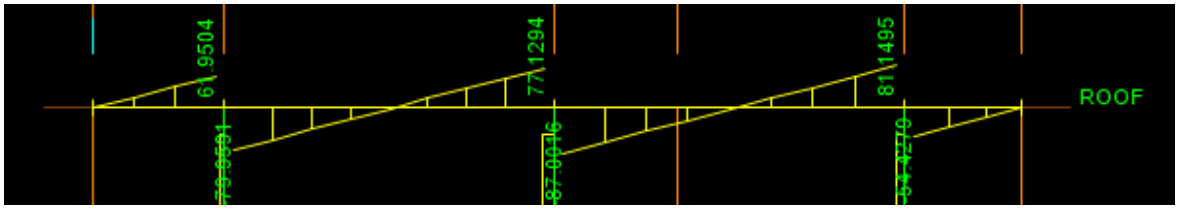
4.1 Design constant

- Cover=25mm
- TTB=30*30
- Longitudinal bar diameter 16 mm and Stirrup 8 mm diameter
- C 20/25
- Fyk=400
- Fck=20mpa
- Fcd= (0.85*Fck)/1.5

  =11.33mPa
- Fyd= Fyk/1.15

  = 400/1.15= 347.83mpa





4.1.1 Depth determination

- P=1.5% for beam
- Po= ($\sqrt{Fck}$)/1000

= ($\sqrt{20}$)/1000

= 0,00447:

- p>po
- L/d=k{11+1.5$\sqrt{Fck}$ po/p + 0}
- Where k=0.4…… for cantiliever

  K=1.3 …… for end

  K=1.5 …… for interior
- For cantiliever

  L/d=k{11+1.5$\sqrt{20}$ po/p + 0}

=0.4{11+1.5$\sqrt{20}$ 0.4472/1.5 + 0}

=5.19998

- Corrected for cantiliever

  L/d*500/Fyk

  =5.19998*500/400 = 6.499975

- For interior

  L/d= k{11+1.5$\sqrt{Fck}$ po/p + 0}

  =1.5{11+1.5$\sqrt{20}$ 0.4472/1.5+ 0}

  =19.4999

- Corrected for interior

  L/d*500/Fyk

  =19.4999*500/400

  =24.3749

$$D_{ye} = \frac{L_{YE}}{L/d_{corr}} = \frac{1500}{6.499975}$$

=230.3749

- $$D_{BX} = \frac{L_{BX}}{L/d_{corr}} = \frac{1330}{6.499975}$$

=204.616

$$D_{BD} = \frac{L_{BD}}{L/d_{corr}} = \frac{4000}{24.3749}$$

=164.103

$$D_{DE} = \frac{L_{DE}}{L/d_{corr}} = \frac{3750}{24.3749} = 153.84678$$

- Take dmax =230.77
- D=25+16/2+8+230.77=271.77mm

  Take D=300mm

- $d_{revised}$ =300-25-8-8=259mm

Check whether T – section and Γ –section or rectangular beam.

be <,bw+le$\sqrt{5}$ = 300+400 $\sqrt{5}$ = 1100

Effective width (be) = 4000mm Using design chart

U= Msd/ Fcdbd^2

$= 51.7556*10^6/ (11.33*300*259^2)$

=0.22698

From The chart Kx. =0.08

Kz= 0.97

Depth of Neutral Axis, X = K x*d = 0.08*259 = 20.72 < hf = 160mm

The neutral axis is found within the flange so the section treated as Rectangular beam of width, bw.

4.2 Design for support beam

- ✓ On support E
  - Msd =43.3653
  - d=259
  - bw=300
  - Nsd=$\frac{msd}{fcdbd^2}$

    $=\frac{43.3653*10^6}{11.33*300*259^2}$

    =0.1901917

step.2.

- Nsd lim =0.295 for 0% redistribution
- Nsdlim>Nsd……single reinforced

Compression reinforced is not required

Step3.from the design table c 20/25

- By using Interpolation…..w=0.2132493

Step 4. Determine area of tension reinforcement

AS_1=1/fyd{w*b*d*fcd*Nsd

$$= \frac{0.2132493*300*259*11.33}{347.83}$$

=539.724

Step5.cheak minimum, max area

$$A_{s.min} = \text{Max} \begin{cases} \frac{0.26*fctm*b*d}{fyk} = (0.26*2.21*300*259)/400 = 111.616mm^2 \\ 0.0013*b*d = 0.0013*300*259 = 101.01mm^2 \end{cases}$$

As.max =0.04*Ac=0.04*300mm*259mm=3108mm

- ❖ Therefore…. Asmin <As< Asmax {111.616<539.724<3108} …. ok!!!

Step 6. Determine number of bars

n= AS/as as= π*d^2/4

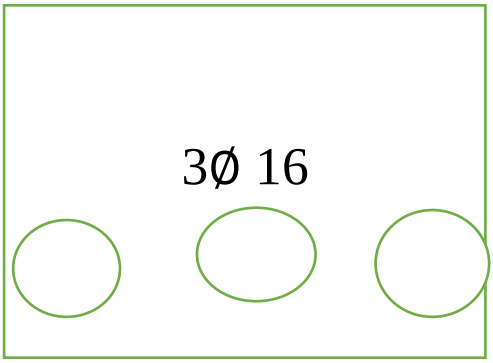
= 3.14*16^2/4

=201.062

=539.724/π*16^2/4

=2.684

provided 3Ø 16



- ✓ ON support D

  Msd=51.7556

  d =259

  bw=300mm

step.1.

- Nsd =msd/fcd*b*d=$\frac{51.7556*1o^6}{11.3*300*259*259}$ = 0.226989963
- Nsdlim = 0.295 …….. 0% redistribution

  Nsdlim>Nsd……single reinforced

Step,2. From the design table c 20/25

Interpolation ……w=0.256508695

Step.3. Determine area of tension reinforcement

As1=$\frac{\{w*b*d*d*fcd}{fyd}$

As1= (0.256508695*300*259*11.33)/347.83

=649.2111699mm^2

Step.5. check minimum, max area

$$A_{s.min}=\text{Max}\begin{cases}\frac{0.26*fctm*b*d}{fyk} & =(0.26*2.21*300*259)/400=111.616mm^2\\ 0.0013*b*d=0.0013*300*259=101.01mm^2\end{cases}$$

As.max = 0.04*Ac = 0.04*300mm*259mm = 3108mm

- ❖ Therefore …. Asmin <As< Asmax {111.616<649.2111699<3108}…..ok!!!

Step 6. Determine number of bars

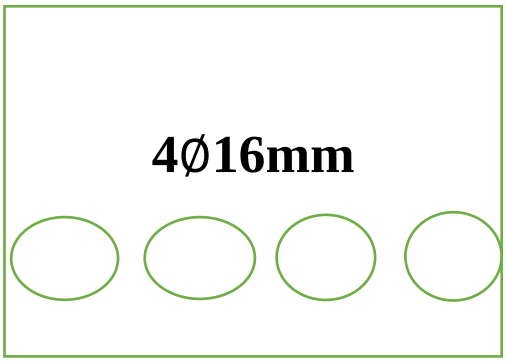
n= AS/as as= π*d^2/4

= 3.14*16^2/4= 201.062

=649.2111699/201.062

=3.2289

Provided **4Ø16mm**



- ✓ ON support B

Msd=33.4731

d =259

bw=300mm

step.1.

- Nsd =msd/fcd*b*d=$\frac{33.4731*1o^6}{11.3*300*259*259}$ = 0.14681
- Nsdlim=0.295 ….. for 0% redistribution

Nsdlim>Nsd……single reinforced

Step,2. From the design table c 20/25

Interpolation ……w=0.161

Step.3. Determine area of tension reinforcement

As1=$\frac{\{w*b*d*d*fcd}{fyd}$

As1= (0.161*300*259*11.33)/347.83

=407.4833mm^2

Step.5. check minimum, max area

$$A_{s.min}=\text{Max}\begin{cases}\frac{0.26*fctm*b*d}{fyk}=(0.26*2.21*300*259)/400=111.616mm^2\\ 0.0013*b*d=0.0013*300*259=101.01mm^2\end{cases}$$

As.max=0.04*Ac=0.04*300mm*259mm=3108mm

- ❖ Therefore….Asmin<As<Asmax{111.616<407.4833<3108}…..ok!!!

Step 6. Determine number of bars

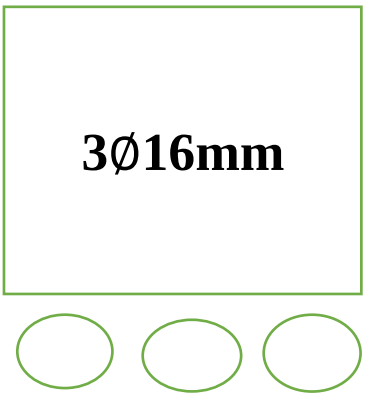
n= AS/as as= π*d^2/4

= 3.14*16^2/4= 201.062

=407.4833/201.062

=2.02665

Provided **3∅16mm**



4.3 Design on Span Beam

- ✓ ON Span ED

Msd=29.175

d =259

bw=300mm

step.1.

- Nsd =msd/fcd*b*d=$\frac{29.175*1o^6}{11.33*300*259*259}$ = 0.12795586
- Nsdlim=0.295 …… for 0% redistribution

Nsdlim>Nsd……single reinforced

Step,2. From the design table c 20/25

Interpolation ……w=0.137547032

Step.3. Determine area of tension reinforcement

As1=$\frac{\{w*b*d*d*fcd}{fyd}$

As1=(0.137547032*300*259*11.33)/347.83=348.12492mm$^{\wedge 2}$

Step.5. check minimum, max area

$$A_{s.min}=Max\begin{cases}\frac{0.26*fctm*b*d}{fyk} & =(0.26*2.2*300*259)/400=111.616mm^2\\ 0.0013*b*d=0.0013*300*259=101.01mm^2\end{cases}$$

As.max=0.04*Ac=0.04*300mm*259mm=3108mm

- ❖ Therefore….Asmin<As<Asmax{111.616<348.1249<3108}…..ok!!!

Step 6. Determine number of bars

n= AS/as as= π*d^2/4

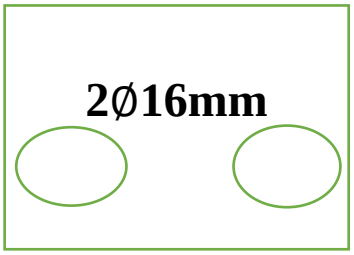
= 3.14*16^2/4

=201.062

=348.1249/201.062

=1.7314

provided 2∅16mm



- ✓ ON Span DB

Msd=33.6757

d =259

bw=300mm

step.1.

- Nsd =msd/fcd*b*d=$\frac{33.6757*1o^6}{11.3*300*259*259}$ = 0.1475635
- Nsdlim=0.295 ……. For 0% redistribution

Nsdlim>Nsd……single reinforced

Step,2. From the design table c 20/25

Interpolation ……w=0.1610762

Step.3. Determine area of tension reinforcement

As1=$\frac{\{w*b*d*d*fcd}{fyd}$

As1=(0.091*300*259*11.33)/347.83=407.67612mm^2

Step.5. check minimum, max area

$$A_{s.min}=\text{Max}\begin{cases}\frac{0.26*fctm*b*d}{fyk} = (0.26*2.2*300*259)/400=111.616mm^2\\ 0.0013*b*d=0.0013*300*259=101.01mm^2\end{cases}$$

As.max=0.04*Ac=0.04*300mm*259mm=3108mm

- ❖ Therefore….Asmin<As<Asmax{111.11<407.6761<3108}…..ok!!!

Step 6. Determine number of bars

n= AS/as as= π*d^2/4

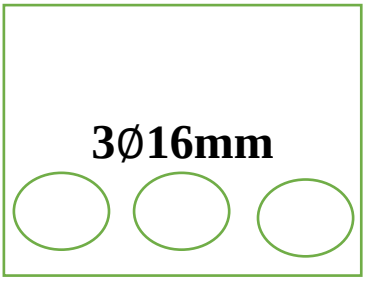
= 3.14*16^2/4

=201.062

=407.6761/201.062

=2.0276

Provided **3∅16mm**



VRDmax =cot ∅ =2.5

VRDmax= (bw*z*v*Fcd)

cot ∅ +tan∅

V=0.6{1-Fck/250}

=0.6{1-20/250}

=0.552

Z=0.9D

=237.6

VRDmax= (300*237.6*0.552*11.33)

2.5 +1/2.5

=445796.523/2.9

=153.72294KN

- **Max shear =87.0016 …… from diagram**
- **VRDmaz >Vmax …… so cot ∅ =2.5**
- $V_{Rd,c} = [C_{Rd,c}k(100\ \rho_l f_{ck})^{1/3} + k_1\ \sigma_{cp}]\ b_w d$

CRdc =0.18/1.5

=0.12

- ***P***1=As/bwd

  ={4π*d^2/4}/(300*259)

  =0.010351

- K=1+ $\sqrt{200/d}$

= 1+ $\sqrt{200/259}$

=1.87875

- VRdc= {0.12*1.87875*$\sqrt[3]{(100 * 0.010351 * 20)}$*300*259

  =48.09966KN

- Vmin = 0.035*K$^{1.5}$*$\sqrt{Fck}$*bwd

  = 0.035*1.87875$^{1.5}$*$\sqrt{20}$*300*259

  =31.31898KN

- VRdc > Vmin ……..ok

Asw/s =Ved/(zFydcot∅)

=87.0016KN/(237.6*347.83*2.5)

=87.0016KN/206.611

=0.421089

- Check $p_{.min}$

  = (0.08*$Fck^{0.5}$)/Fyk

  = ($0.0820^{0.5}$)/400

  =0.000894

- $Asw_{.min}$ / s = $pw_{.min}$ *bwsin α α=1 ….. it is vertical stirrup

  =*0*.000894*300*1

  =*0*.2682mm2

$Asw_{.min}$ / s < $Asw_{.}$/ s ……..ok

Asw = (4π*d^2)/4

=(4π*8^2)/4

= 201.061

- Spacing

  =0.75d

  =0.75*259

  =194.25 …….. provided 190

Provide ∅8 c/c 190

Chapter 5

COLUMN DESIGN

5.1. Introduction

Column is a vertical structural member supporting axial compressive load with or without moment. The cross-sectional dimensions of column are generally considerably less than its height. It supports vertical loads from the floors and roof and transmit this load to foundation.

10.2 Classification of Columns

Based on lateral Reinforcement

i. Tied columns: where the main longitudinal bars are enclosed within closely spaced *lateral ties*,
ii. Spiral columns: where the main longitudinal bars are enclosed within closely spaced and continuously wound spiral reinforcement.

Based on nature of Loading

o Columns with axial loading (applied concentrically)

o Columns with uniaxial eccentric loading

o Columns with biaxial eccentric loading

Based on Slenderness Ratio

Slenderness‟ is a geometrical property of a compression member which is related to the ratio of its „effective length‟ to its lateral dimension. Also it provides a measure of the vulnerability to failure of the column by elastic instability (buckling) in the plane in which the slenderness ratio is computed.

Generally depending on whether slenderness effects are considered insignificant or not:

i. Short columns is column with low slenderness ratios, i.e. it is relatively short and stocky columns, invariably fail under ultimate loads with the material (concrete, steel; reaching its ultimate strength, and not by buckling

m ii. Slender (long) columns is Columns with very high slenderness ratios in danger of buckling accompanied with large lateral deflection under relatively low compressive loads, and thereby failing suddenly. Here columns buckle and the additional moments caused by deflection must be taken into account in design. Design codes attempt to preclude such failure by specifying „slenderness limits‟ to columns.

Stability effects must be considered in the design of compression members.

10.3 Analysis and design Procedures

The internal forces and moments may generally be determined by elastic global analysis using either: -First-order theory, using the initial geometry of the structure, or Second-order theory taking in to account the influence of the deformation of the structure.

First-order theory maybe used for the global analysis in the following cases:

(a) Non-sway frames
(b) Braced frames
(c) Design methods which make indirect allowances for second-order effects.

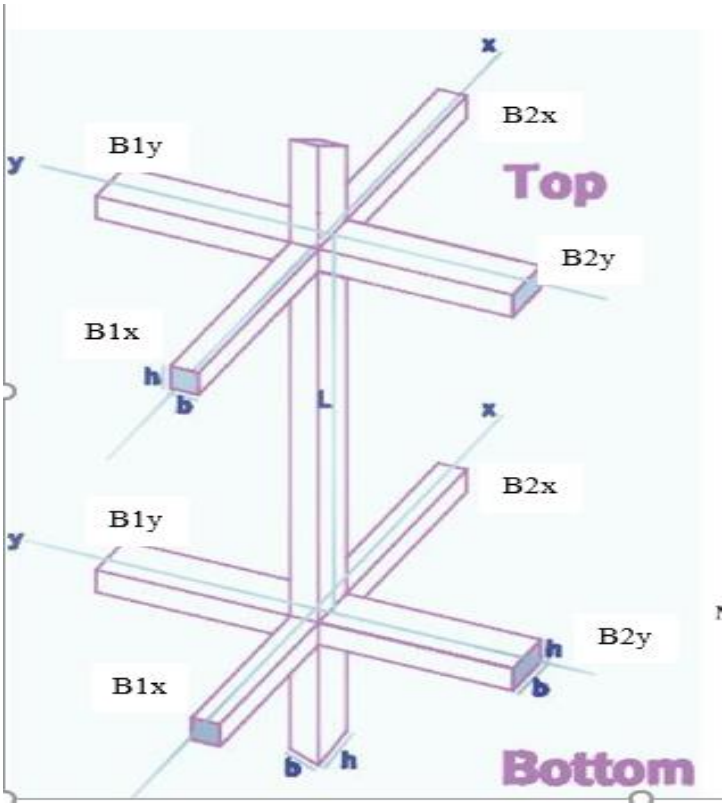
 Second-order theory maybe used for the global analysis in all cases. Design for structural stability taking account of second-order effects shall ensure that, for the most Unfavorable combinations of actions at the ultimate limit state ,loss of static equilibrium (locally or for the structure as a whole) does not occur or the resistance of individual cross-sections subjected to bending and longitudinal force is not exceeded.

This thesis project was conduct the axial load and moment from etabs output in three combinations listed above to help us to find the effect of of axial force plus moment.

Our column is bi axial column from first floor to second floor
Material property C40X40 and beam cross section B25X40
Fyk= 400mpa steel grade

Concrete data	
fcu(Mpa)=	C20/25
gmc =	1.5
Ecm(Gpa)=	30
cover (mm)=	30
Reinforcement data	
fyk(Mpa)=	400
gms =	1.15
Es(Gpa)=	200
main barø(mm)	16
tie barø(mm)	8

Design based on Nominal stiffenes



FROM ETABS WE GET THE FOLLOWING VALUES:

X-SECTIONAL DATA			
Element	Depth 'h' (Y-dxn) (mm)	width 'b' (X-dxn)(mm)	length 'L' (mm)
column	300	200	2880
Top beam			
B1 (x-x) (pin)	300	300	4000
B2 (x-x) (continous)	300	300	5000
B1 (y-y) (cantilever)	300	300	1330
B2 (y-y) (continous)	300	300	4000
Bott beam			
B1 (x-x) (pin)	400	300	4000
B2 (x-x) (continous)	400	300	5000
B1 (y-y) (cantilever)	400	300	1330
B2 (y-y) (continous)	400	300	4000

AND DESIGN ACTIONS

design actions	
N_{ED}(KN)=	470.4866
M_{x-x}, T(KN-M)=	-7.401
M_{x-x}, B(KN-M)=	10.45
M_{y-y}, T(KN-M)=	-16.1057
M_{y-y}, B(KN-M)=	22.0747

- ***Material property***

$$C25/30 = f_{cd} = \frac{0.85*25}{1.5} = 14.167 Mp_a$$

$$f_{ctm} = 2.6 Mp_a$$

$$S400 \quad f_{yd} = \frac{400}{1.15} = 347.826 Mp_a$$

E_{cm}= 31 Gpa

E_s= 200 Gpa

ϵ_{yd}= 1.74‰

Design producers

Step 1:-Determine material design value

$fcd = \alpha cc\ fc\ /\gamma c$ = 0.8*25/1.5= 14.17N/ m2

$fyd = fy$k/ γs = 400/1.15 = 347.83N/mm2

Step 2 : determination of effective length.

Effective length

For braced member

$$lo = 0.5l\sqrt{\left[1+\frac{k_1}{0.5+k_1}\right]*\left[1+\frac{k_2}{0.5+k_2}\right]}$$

But

$$Ki = \frac{(\frac{EI}{l})column}{\sum(\frac{2EI}{l})beam}$$

So we can calculate moment of inertia.

For first floor beam

$I = \frac{BH^3}{12}$ =300$x\frac{40\dot{O}^3}{12}$=1600000000 mm^4

For top tile beam

$I = \frac{BH^3}{12}$ =300$x\frac{30\dot{O}^3}{12}$=675000000 mm^4

For column x-x=$\frac{HB^3}{12}$=300*200^3/12=200000000mm^4

For column y-y=$\frac{BH^3}{12}$=200*300^3/12=450000000mm^4

For K1(xx)at the top =$\frac{\frac{Icxx}{Lc1}+\frac{Icxx}{Lc2}}{2\left[\frac{1}{2}\left(\frac{Ib1}{Lb1}\right)+\frac{Ib2}{Ib2}\right]}$

where for pin the beam must $\frac{1}{2}\left(\frac{Ib1}{Lb1}\right)$

$$=\frac{\left(\frac{200000000}{2880}\right)*2}{\left(\frac{675000000}{4000}(\frac{1}{2})+\frac{675000000}{5000}\right)*2}$$

$$=\frac{69444.444}{84375+135000}$$

K1(xx)=0.317

K2(xx) ON BOTTOM

where for pin the beam must $\frac{1}{2}\left(\frac{Ib1}{Lb1}\right)$

$$=\frac{\left(\frac{200000000}{2880}\right)*2}{\left(\frac{1600000000}{4000}(\frac{1}{2})+\frac{1600000000}{5000}\right)*2}$$

$$=\frac{69444.444}{200000+320000}$$

K1(xx)=0.134

For K1(yy) at the top $=\frac{\frac{Icyy}{Lc1}+\frac{Icyy}{Lc2}}{2\left[\left(\frac{Ib1}{Lb1}\right)+\frac{Ib2}{Ib2}\right]}$

$$=\frac{\left(\frac{450000000}{2880}\right)*2}{\left(0+\frac{67500000000}{4000}\right)*2}$$

$$=\frac{156750}{0+168750}$$

= 0.926

For K2(yy) bottom

$$=\frac{\frac{Icyy}{Lc1}+\frac{Icyy}{Lc2}}{2\left[\left(\frac{Ib1}{Lb1}\right)+\frac{Ib2}{Ib2}\right]}$$

$$=\frac{\left(\frac{450000000}{2880}\right)*2}{\left(0+\frac{1600000000}{4000}\right)*2}$$

$$=\frac{156250}{0+400000}$$

= 0.391

Then we can find effective length

$$Lo = 0.5L\sqrt{\left(1+\frac{K1}{0.45+K1}\right)\left(1+\frac{K2}{0.45+K2}\right)}$$

$$Lox(mm)= 0.5 * 2880\sqrt{\left(1+\frac{0.317}{0.5+0.317}\right)\left(1+\frac{0.134}{0.5+0.134}\right)}$$

$$=0.5 * 2880\sqrt{1.388 * 1.21}$$

Lox(mm)= 1866.55

$$Lo = 0.5L\sqrt{\left(1+\frac{K1}{0.45+K1}\right)\left(1+\frac{K2}{0.45+K2}\right)}$$

$$Loy(mm)= 0.5 * 2880\sqrt{\left(1+\frac{0.926}{0.5+0.926}\right)\left(1+\frac{0.391}{0.5+0.391}\right)}$$

$$=0.5 * 2880\sqrt{1.649 * 1.439}$$

Loy(mm)=2218.14

Step 3 Cheek Minimum reinforcement

$$Asmin=\max\begin{Bmatrix}0.1 * \frac{Ned}{fyd}\\ 0.002Ac\end{Bmatrix}$$

$$=\begin{Bmatrix}0.1 * \frac{470.4865}{347.83} = 135.26\\ 0.002 * 200 * 300 = 120\end{Bmatrix}$$

where 470.4865 is the axial load which obtain from etabs.

Asmin =135.26mm^2

Cheek Maximum reinforcement

Asmax =0.04Ac

=0.04*200*300

Asmax =2400mm^2

Then Asmin <Asprovide<Asmax it is safe.

Step 4 design actions
Imperfection eccentricity

$$E=\max\left\{\begin{matrix}\frac{Lo}{400}\\ \frac{H}{30}\\ 20\end{matrix}\right\}$$

$$e1(xx)=\max\left\{\begin{matrix}\frac{Lox}{400}\\ \frac{H}{30}\\ 20\end{matrix}\right\}$$

$$e1(xx)=\max\left\{\begin{matrix}\frac{1866.85}{400}=4.666375\\ \frac{300}{30}=10\\ 20\end{matrix}\right\}$$

e1(xx)=20

$$e1(yy)=\max\left\{\begin{matrix}\frac{Loy}{400}\\ \frac{H}{30}\\ 20\end{matrix}\right\}$$

$$e1(yy)=\max\left\{\begin{matrix}\frac{2218.14}{400}=5.545\\ \frac{300}{30}=10\\ 20\end{matrix}\right\}$$

e1(yy)=20

Step 5 Cheek slenderness.

$$\lambda=\frac{l_o}{i}\ i=\sqrt{\frac{I}{A}}\ i$$

In the x direction

$$i=\sqrt{\frac{Icxx}{A}}$$

$$=\sqrt{\frac{200000000}{200*300}}$$

=57.735

$$\lambda=\frac{Lox}{i}$$

$$=\frac{1866.55}{57.735}$$

=32.329

In the y direction

$$i=\sqrt{\frac{Icxx}{A}}$$

$$=\sqrt{\frac{450000000}{200*300}}$$

=86.6

$$\lambda=\frac{Loy}{i}$$

$$=\frac{2218.14}{86.6}$$

=25.61

First order moment

Along xx

Mo1=min$\begin{Bmatrix} mx_x top \\ mx_x bottom \end{Bmatrix}$

=min$\begin{Bmatrix} -7.401 \\ 10.45 \end{Bmatrix}$

=7.454

Mo2= max$\begin{Bmatrix} mx_x top \\ mx_x bottom \end{Bmatrix}$

=max$\begin{Bmatrix} -7.401 \\ 10.45 \end{Bmatrix}$

=10.45

Along yy

Mo1=min$\begin{Bmatrix} my_y\ top \\ my_y\ bottom \end{Bmatrix}$

=min$\begin{Bmatrix} -16.1057 \\ 22.0747 \end{Bmatrix}$

=16.1057

Mo2=max$\begin{Bmatrix} my_y top \\ my_y bottom \end{Bmatrix}$

=max$\begin{Bmatrix} -16.1057 \\ 22.0747 \end{Bmatrix}$

=22.0747

Along xx

Moe=max$\begin{Bmatrix} -0.4mo1 + 0.6mo2 \\ 0.4mo2 \end{Bmatrix}$

= max$\begin{Bmatrix} -0.4 * 7.401 + 0.6 * 10.45 \\ 0.4 * 10.45 \end{Bmatrix}$

=max$\begin{Bmatrix} 3.3096 \\ 4.18 \end{Bmatrix}$

=4.18

Along yy

Moe=max$\begin{Bmatrix} -0.4mo1 + 0.6mo2 \\ 0.4mo2 \end{Bmatrix}$

= max$\begin{Bmatrix} -0.4 * 16.1057 + 0.6 * 22.0747 \\ 0.4 * 22.0747 \end{Bmatrix}$

=max$\begin{Bmatrix} 6.8 \\ 8.8299 \end{Bmatrix}$

=8.8299

Step 6 calculate slenderness limit

$\lambda \lim = \frac{20ABC}{\sqrt{n}}$ where A=0.7,B=1.1,C=1.7-rm

Along xx

C=1.7-rm

Cxx=1.7-$\left\{\frac{-mo1}{mo2}\right\}$

=1.7-$\left\{\frac{-7.401}{10.45}\right\}$

=2.408

Along yy

C=1.7-rm

Cyy=1.7-$\left\{\frac{-mo1}{mo2}\right\}$

=1.7-$\left\{\frac{-16.1057}{22.0747}\right\}$

=2.4296

$$n = \frac{N_{ed}}{Acf_{cd}} = \frac{470.4866*10^3}{200*300*11.33}$$

=0.692096

$$\lambda \lim = \frac{20ABC}{\sqrt{n}}$$

$$=\frac{20*0.7*1.1*2.408}{\sqrt{0.692096}}$$

=44.5753>25.53 therefor short column along xx

$$\lambda \lim = \frac{20ABC}{\sqrt{n}}$$

$$=\frac{20*0.7*1.1*2.4296}{\sqrt{0.692096}}$$

=44.97512>24.285 therefor short column and neglect second order effect.

Step 7 Design moments

Along yy

Msdy= max$\left\{\begin{matrix} mo2(yy) \\ moe + e2 + \frac{eyy*Ned}{1000} \\ mo1 + 0.5e2 \end{matrix}\right\}$ neglect e2

=max$\left\{\begin{matrix} 22.0747 \\ 8.8299 + 0 + \frac{20*470.4865}{1000} = 18.23963 \\ 16.1057 + 0 = 16.1057 \end{matrix}\right\}$

Msdy=22.0747

Along xx

Msdx= max$\left\{\begin{matrix} mo2(xx) \\ moe + e2 + \frac{exx*Ned}{1000} \\ mo1 + 0.5e2 \end{matrix}\right\}$ neglect e2

$$=\max\left\{\begin{matrix} 10.45 \\ 4.18+0+\frac{20*470.4865}{1000}=13.58973 \\ 7.401+0=7.401 \end{matrix}\right\}$$

Msdx=13.58973

Step 8 main reinforcement

Vsd=$\frac{Ned}{Fcd*Ac}$

$$=\frac{470.4865*10^3}{11.33*200*(300-41)}$$

$$=0.8017\ where\ D=\left(300-25-\frac{16}{2}-80\right)=41$$

Nedx=$\frac{\mathrm{msdx}}{fcdbd^2}$

$$=\frac{13.58973*10^6}{11.33*200*300^2}$$

=0.0666

Nedy=$\frac{\mathrm{msdy}}{fcdbd^2}$

$$=\frac{22.0747*10^6}{11.33*200*300^2}$$

=0.108241

Step 9 Calculate no of bar

n=$\frac{As}{as}$ $where\ as=\pi d^2/4$

$$=3.14*\frac{16^2}{4}=200$$

n=$\frac{195.5}{200}=0.97\approx 1$

provided 4 Φ16

step 10 calculation of stirrup spacing

we use a diameter of 8mm stirrup

spacing $=\min\begin{Bmatrix} 20 * \Phi 16 \\ smallest\ column\ dimention \\ 400 \end{Bmatrix}$

$=\min\begin{Bmatrix} 20 * 16 = 320 \\ 200 \\ 400 \end{Bmatrix}$

Provide $\Phi 8$ c/c 200mm

CHAPTER 9

FOUNDATION DESIGN

Design of Isolated Footing

An isolated footing is a footing that carries a single column. The function of an isolated footing is to spread the column load laterally to the soil so that the stress intensity is reduced to a value that the soil can safely carry. The approximate contact pressure under a given symmetrical foundation can be obtained from the flexural formula, provided that the considered load lies within the kern of the footing.

Depth of footing:

According to EBCS-7-1995:2015

- ✓ Minimum depth of footing should be 50cm
- ✓ For footings on sloping sites, minimum depth of footing should be 60cm and 90cm below ground surface on rocky and soil formations, respectively.

Footing at different elevations:

- ✓ When adjacent footings are to be placed at different levels, the distance between the edges of footings shall be such as to prevent undesirable overlapping of stresses in soils and disturbance of the soil under the higher footing due to excavation for the lower footing. A minimum clear distance of half the width of the footing is recommended.

Material used

- C20/25
- Cover 50mm
- Fyk =400
- Φ=14

ETABS OUTPUT :

Story	Joint	Load combo	Fz	Mx	My
Base	**C14**	**ULS**	**4090.3148**	**0.9881**	**16.1792**
Base	**C14**	**SLS**	**1726.1173**	**0.2977**	**6.9364**

To start the calculation first let's calculate the ultimate bearing capacity of the soil

Data collected from geotechnical for soil test results

.

Soil parameters:

C=19kpa, f =17deg and unit weight of 17KN/m^3

Determination of factor of safety

$$F.S = \frac{Ultimate\,\text{load}}{\text{Service load}} =$$

$$\frac{4090.3148+0.9881+16.1792}{1726.1173+0.2997+6.9364} = 2.37$$

Determination of Ultimate bearing capacity of soil using Hansen's equation

The Hansen equation is given in Table 4-1 as

$$q_{ult} = cN_c s_c d_c + \bar{q}N_q s_q d_q + 0.5\gamma B' N_\gamma s_\gamma d_\gamma$$

Assume Square Footing

From table for f = 17 deg

Nc = 11 Nq = 4 Ng = 1.2

Sc= 1.3 Sq=1.2 Sg=0.8

dc = 1+0.35(D/B) dq = 1+0.35(D/B) dg=1 Where D=2.5m

$q_{u=}$ 19×11×1.3×(1+0.35)2.5/B + 17×4×0.8×(1+0.35)2.5/B +0.5×17×B×1.2×0.8×1

$=\frac{1100.58}{B} + 81.63B$

$A \times q_{ult} = P \times F_s$

$$A = \frac{P \times F_s}{Q_{ult}}$$

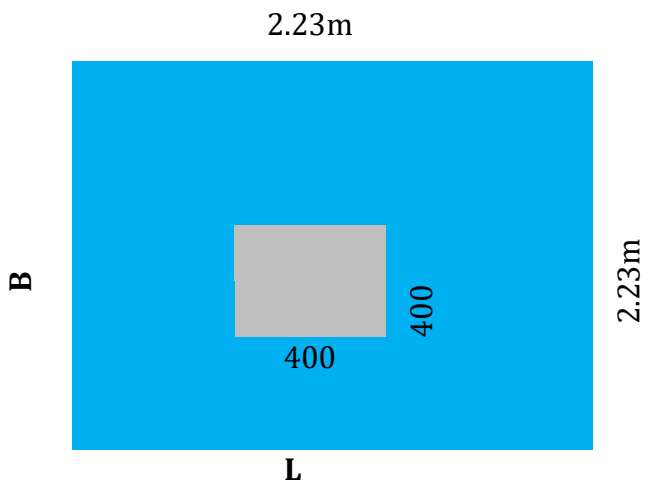
$$B^2 = \frac{4090.3148 * 2.369}{1853.69 + 42.341B + 946.361}$$

B

By Trial and Error, Footing Dimension = **2.23×2.23**

Q_u = 483.96 + 96.9 + 13.056 = **1849.626Kpa**

The Allowable Bearing Capacity = Q_{ult} /F.s = 1849.6975/2.369 = **780.626Kpa**



Step 2 : Eccentricity at Service load

$$e_x = \frac{M_y}{P} =$$

6.9364/ 1726.1173 = 0.0040185

$$e_y = \frac{M_x}{P} =$$

0.2997/1726.1173 = 0.00017247

Step 3: Determination of Stress at four corners

$$\sigma_{max} = \frac{P}{A}(1 + \frac{6e_x}{B} + \frac{6e_y}{B}) \leq \sigma_{all}$$

$\sigma_1 = \frac{1726.1173}{2.23*2.23}(1+\frac{6(0.0040185)}{2.23}+\frac{6(0.00017247)}{2.23})$ = **351.02Kpa**

$\sigma_2 = \frac{1726.1173}{2.23*2.23}(1+\frac{6(0.0040185)}{2.23} - \frac{6(0.00017247)}{2.23})$ = 350.69Kpa

$\sigma_3 = \frac{1726.1173}{2.23*2.23}(1-\frac{6(0.0040185)}{2.23} + \frac{6(0.00017247)}{2.23})$ = 343.51Kpa

$\sigma_4 = \frac{1726.1173}{2.23*2.23}(1-\frac{6(0.0040185)}{2.23} - \frac{6(0.00017247)}{2.23})$ = 343.19Kpa

σ_{max} < Allowable Bearing Capacity

351.02 < 780.626 Kpa……OK!

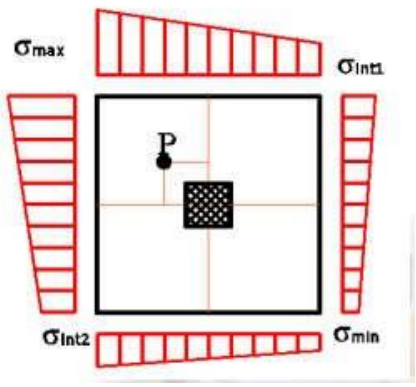
Step 4: Eccentricity at ultimate load

$e_x = \frac{M_y}{P} =$ 16.1792/4090.3148 = 0.0039554

$e_y = \frac{M_x}{P} =$ 0.988/4090.3148 = 0.000242

Step 5: Determination of stress at four corners

$$\sigma = \frac{P}{A}(1 \pm \frac{6e_x}{B} \pm \frac{6e_y}{B})$$



$\sigma_1 = \frac{4090.3148}{2.23*2.23}(1+\frac{6(0.0039554)}{2.23}+\frac{6(0.000242)}{2.23})$ = **831.79Kpa**

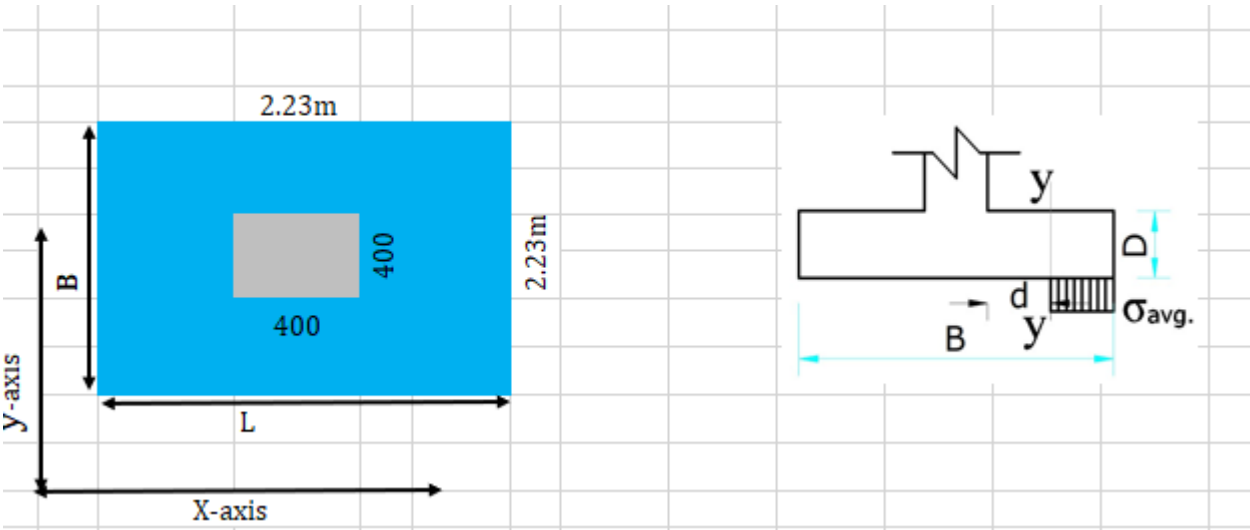
$\sigma_2 = \frac{4090.3148}{2.23*2.23}(1+\frac{6(0.0039554)}{2.23}-\frac{6(0.000242)}{2.23})$ = **831Kpa**

$\sigma_3 = \frac{4090.3148}{2.23*2.23}(1-\frac{6(0.0039554)}{2.23}+\frac{6(0.000242)}{2.23})$ = **817.44Kpa**

$\sigma_4 = \frac{4090.3148}{2.23*2.23}\left(1-\frac{6(0.0039554)}{2.23}-\frac{6(0.000242)}{2.23}\right)$ = **809.81Kpa**

$\sigma_{avg} = \frac{\sigma_1 + \sigma_2 + \sigma_3 + \sigma_4}{4}$ = 822.51Kpa

$\sigma_{avg} < \sigma_{ult}$ **.....OK!**



Step 6 : Check thickness

Check for punching shear

1.The punching resistance of **column bases should be** verified at control perimeters **within** 2d from the periphery of the column [ES EN 1992-1-:2105. Article 6.4.4(2)].

2 Control perimeters at a **distance less than 2d** should be considered where the concentrated force is opposed by a high pressure (e.g. soil pressure on a base) [ES EN 1992-1-:2105.Article 6.4.2(2)].

Assuming footing pad thickness = 0.4

d= D- cover- Φ/2 = 700-50-16/2 = 632

$V_{sd} = \sigma_{avg} \times B\left(\frac{B}{2} - \frac{H}{2} - d\right)$ = 822.51 ×2.23(2.23/2 – 0.4/2 – 632) = 335.663

$V_{Ed,red} = V_{Ed} - \Delta V_{Ed}$ [ES EN 1992 – 1 – 1: 2105, Eqn. 6.48]

$V_{Ed} = \frac{V_{Ed,red}}{ud}$ [ES EN 1992 – 1 – 1: 2105: Eqn. 6.49

$V_{Ed,red}$ = Pd–[σavg (C x)*(C y)]- (ΔVEd)

= 4090.3148 – (822.51×0.4×0.4) – 396.73= 2109.862KN

Punching Shear Capacity of Footing

$$V_{rd,max} = \frac{\alpha_{cw} u_o z v_1 f_{cd}}{\cot(\theta) + \tan(\theta)}$$ [ES EN 1992 – 1 – 1: 2015, Eqn. 6.9(with b replaced with u)]

$$v_1 = v = 0.6\left[1 - \frac{f_{ck}}{250}\right] =$$

$$u = 2 * (3d + c_x) + 2 * (3d + c_y)$$

U_0 = 2×(0.4×0.4) = 1.6m

Z=0.9d = 0.9×742 = 667.8

V=0.6(1-20/250) = 0.552

θ = 22, Cotθ = 2.5, tanθ = 0.4

$\frac{1.6\times667.8\times0.552\times11.33}{2.5+0.4}$ = 2304.29KN > V_{sd} …..Punching shear is safe at column face!!

Check Punching Shear At 1.5d from face of column

$$V_{rd,c} = C_{rd,c} k (100\rho_1 f_{ck})^{\frac{1}{3}} ud \geq V_{min}\, ud$$ [ES EN 1992 – 1 – 1: 2015, Eqn. 6.47(with b = u)]

$$C_{rd,c} = \frac{0.18}{\gamma_c} = \frac{0.18}{1.5} = 0.12$$

$$\rho_1 = \rho_{min} = 0.26\frac{f_{ctm}}{f_{yk}} = 0.26 * \frac{0.3 f_{ck}^{\frac{2}{3}}}{f_{yk}} =$$

$$u = 2 * (3d + c_x) + 2 * (3d + c_y)$$

$$k = 1 + \sqrt{\frac{200}{d}} \leq 2$$

$$V_{rd,c} = (v_{min})ud$$ [ES EN 1992 – 1 – 1: 2015, Eqn. 6.2 b(with U = b)]

$$v_{min} = 0.035 k^{\frac{3}{2}} f_{ck}^{\frac{1}{2}}$$ [ES EN 1992 – 1 – 1: 2015, Eqn. 6.3 N]

u = 2(3×0.782+0.4) + 2(3×0.782+0.4)

= 10.984m

ρ_{min} = 0.26×0.3×$20^{2/3}$ /400 = 0.0014

k= 1+$\sqrt{\frac{200}{742}}$ = 1.5625

V_{rdc} = $0.12 \times 1.5625 \times (100 \times 0.0014 \times 20)^{1/3} \times 10.984 \times 782$ = 22699.97

V_{min} =$0.035 \times 1.5625^{\frac{3}{2}} \times 20^{\frac{1}{2}} \times 10.984 \times 343$ = **2364.720.KN** > **V_{sd}** …….OK!! Safe at 1.5d

Step 7: Wide beam Shear

According to ES EN 1992-1-1:2015, Article 6.2.1(8) the Critical section for shear is at distance d from the face of supports

$$V_{rd,c} = \left[C_{rd,c}k(100\rho_1 f_{ck})^{\frac{1}{3}}\right] b_w d \qquad \text{[ES EN 1992 – 1 – 1: 2015, Eqn. 6.2 a]}$$

$$C_{rd,c} = \frac{0.18}{\gamma_c} = \frac{0.18}{1.5} = 0.12$$

$$\rho_1 = \rho_{min} = 0.26\frac{f_{ctm}}{f_{yk}} = 0.26 * \frac{0.3f_{ck}^{\frac{2}{3}}}{f_{yk}} =$$

$$V_{rd,c}(\min) = (v_{min})b_w d$$

$$k = 1 + \sqrt{\frac{200}{d}} \leq 2$$

$$v_{min} = 0.035k^{\frac{3}{2}}f_{ck}^{\frac{1}{2}}$$

ρ_{min} = 0.26×0.3×$20^{2/3}$ /400 = 0.0014

k= 1+$\sqrt{\frac{200}{632}}$ = 1.5625

$V_{rd,c}$ = (0.12×1.5625(100×0.0014×20)$^{1/3}$) × 0.4 ×632 = 480.0916KN

$V_{rd,c\ (min)}$ = 0.035×$1.5625^{3/2}$ ×$20^{1/2}$ ×0.4×632 =435.84 **KN > Vsd** ……OK!!

Step 8 : Reinforcement required

Parallel to X-Direction

$M_{sd}x = \sigma_{avg} \times B(\frac{L}{2} - \frac{H}{2})^2$

= 822 × 2.23($\frac{2.23}{2} - \frac{0.4}{2}$)^2

= 609.17KNm

$\mu sd_x = \frac{Msdx \times 10^6}{fcd \times 1000 \times d^2} = \frac{609.17 \times 10^6}{11.33 \times 1000 \times 742^2} = 0.045$

From design table by interpolation

K_z = 0.968

$A_s = \frac{Msdx \times 10^6}{Kz \times d \times fyd} = \frac{609.17 \times 10^6}{0.924 \times 343 \times 742.82} = 1377.64mm^2$

$As_{min} = 0.26 \times \frac{Fctm}{fyk} \times 1000 \times d = 0.26 \times \frac{2.2}{400} \times 1000 \times 742 = 1066.09mm^2$

$As_{max} = 0.04 \times 1000 \times d = 0.04 \times 1000 \times 742 = 29280mm^2$

$As_{min} < A_s < As_{max}$ … OK!! As provided = As = 1377.64mm

$$\text{Spacing} = \begin{cases} \frac{b \times as}{Asprovided} = \frac{1000 \times 3.14 \times 782^2/4}{1377.64} = 140\text{mm} \\ 3d \quad = 3 \times 782 = 2346\text{mm} \\ 400\text{mm} \end{cases}$$

Minimum = 140mm **Provide Φ=16 C/c 140mm**

Parallel to Y- Direction

$M_{sd}y = \sigma_{avg} \times B\,(\frac{L}{2} - \frac{H}{2})^2$

$= 822 \times 2.23(\frac{2.23}{2} - \frac{0.4}{2})^2$

$= 609.17$KNm

$\frac{Msdy}{B} = 301.13/1.6 = 188.208$KNm

$\mu sd_y = \frac{Msdy \times 10^6}{fcd \times 1000 \times d^2} = \frac{609.17 \times 10^6}{11.33 \times 1000 \times 742^2} = 0.045$

From design table by interpolation

$K_z = 0.968$

$A_s = \frac{Msdy \times 10^6}{Kz \times d \times fyd} = \frac{188.208 \times 10^6}{0.924 \times 742 \times 347.82} = 1377.64\text{mm}^2$

$As_{min} = 0.26 \times \frac{Fctm}{fyk} \times 1000 \times d = 0.26 \times \frac{2.2}{400} \times 1000 \times 742 = 1066.09\text{mm}^2$

$As_{max} = 0.04 \times 1000 \times d = 0.04 \times 1000 \times 742 = 29680\text{mm}^2$

$As_{min} < A_s < As_{max}$ … OK!!

Asprovided = As = 1377.64mm^2

$$\text{Spacing} = \begin{cases} \dfrac{b \times as}{Asprovided} = \dfrac{1000 \times 3.14 \times d^2/4}{1707.3} = 140\text{mm} \\ 3d \quad = 3 \times 742 = 2226\text{mm} \\ 400\text{mm} \end{cases}$$

Minimum = 140mm **Provide Φ=16 C/c 140mm**

RECOMMENDATION

- The overall project works of the building should be inspected and supervised throughout the entire construction time in order to achieve the required design strength and be sure is that should be serve for its intended purpose.

- The quality of material should fulfil the specification given in the else there may be failure due to using less qualified material.

- Care should be taken when handling, casting and placing structural element especially support condition on construction time must be the same as that of previously determined arrangement on design part.